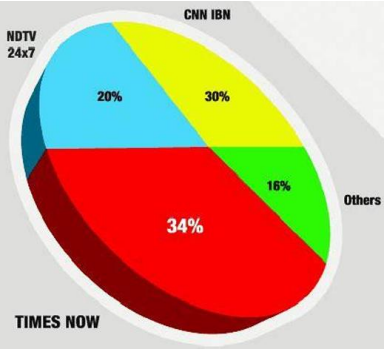
4 – Misleading Visualizations

Common Mistakes, Case studies

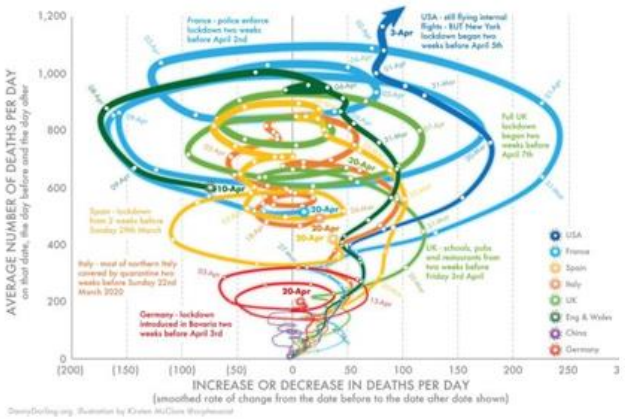
Misleading Visualizations

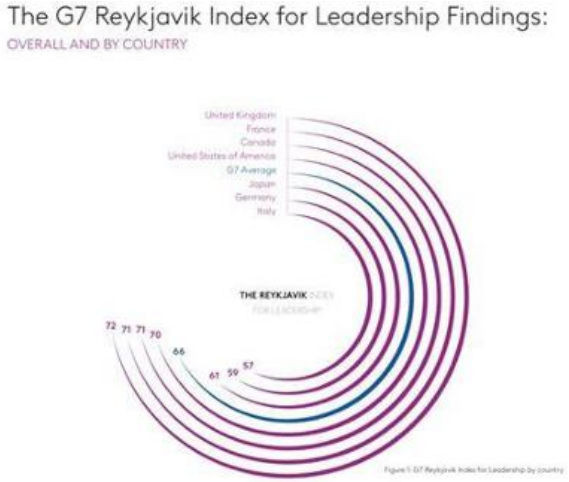
- **Misinformative Visualization**: reader gets a different message than what is in the visualization

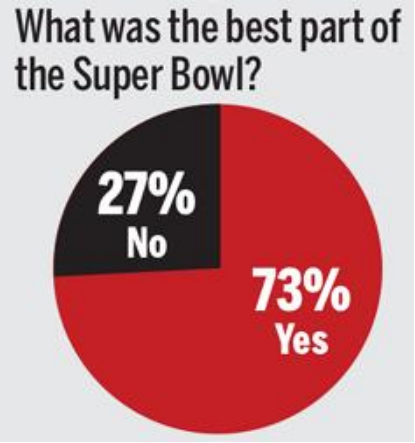


- **Uninformative Visualization**: reader cannot get any information from the visualization

Sometimes the information *is* there, it just takes a while to understand what is going on







Common Visualization Mistakes

- **Input**
  - Garbage in, Garbage out
  - Lie with Statistics
- **Visualization Design**
  - Choice of Axis
  - Choice of Chart
  - Color Mess
- **Plotting**
  - Incomplete Chart
  - Inconsistencies
  - Chaotic Canvas
- **Perception**
  - Visual Illusion
- **Interpretation**
  - False Linkage
  - Power of Words
  - Hidden Information

Garbage-in (152)	Not Data (50)
	Selective Data (40)
	Dubious Data (37)
	Non Sequitur (24)
	Too Few Data Points (1)
Lie with Statistics (110)	Discretized Continuous Variable (28)
	Missing Normalization (26)
	Inappropriate Item Order (23)
	Inappropriate Metric (9)
	Questionable Prediction (9)
	Trend Line on Random Data (7)
	Inappropriate Use of Accumulation (5)
	Inappropriate Aggregation Granularity (2)
	Two-Way Normalization (1)

False Linkage (28)	Invalid Comparison (16)
	Correlation Not Causation (7)
	Pattern Seeking (5)
Power of Words (28)	Misleading Claim (17)
	Misleading Annotation (7)
	Misleading Title (3)
	Misleading Value Labels (1)
Hidden Information (22)	Hidden Distribution (9)
	Overplotting (8)
	Hidden Uncertainty (4)
	Hidden Population Size (1)

Visual Illusion (198)	3D (105)
	Area Encoding (35)
	Ineffective Color Scheme (28)
	Pictorial Area Encoding (19)
	Inappropriate Use of Smoothing (4)
	Distractive Value Labels (3)
	Map Projection Distortion (2)
	Inappropriate Aspect Ratio (1)
	Sine Illusion (1)

Choice of Axis (324)	Truncated Axis (171)
	Dual Axis (78)
	Inappropriate Axis Range (32)
	Inverted Axis (19)
	Log Scale (9)
	Extended Axis (7)
	Data of Different Magnitudes (6)
	Linear Scale on Exponential Data (2)
Choice of Chart (71)	Inappropriate Use of Line Chart (23)
	Inappropriate Use of Pie Chart (17)
	Confusing Chart Type (16)
	Misusing Circular Layout (7)
	Inappropriate Use of Stacked (5)
	Inappropriate Use of Bar Chart (2)
	Inappropriate Use of Scatterplot (1)
Color Mess (64)	Overusing Colors (32)
	Indistinguishable Colors (31)
	Color Blind Unfriendly (1)

Incomplete Chart (286)	Missing Title (94)
	Missing Axis Title (62)
	Missing Legend (52)
	Missing Value Labels (35)
	Missing Axis (25)
	Missing Axis Ticks (17)
	Missing Units (1)
Inconsistency (159)	Misrepresentation (65)
	Inconsistent Tick Intervals (52)
	Inconsistent Binning Size (25)
	Changing Scale (10)
	Violating Color Convention (4)
	Inconsistent Grouping (1)
	Inconsistent Tick Labels (1)
	Inconsistent Value Labels (1)
Chaotic Canvas (80)	Cluttering (19)
	Confusing Legend (16)
	Plotting Error (14)
	Missing Abbreviation (10)
	Plotting Out of Chart (8)
	Misalignment (7)

Don’t memorize; be critical: can this be misinterpreted in some way?

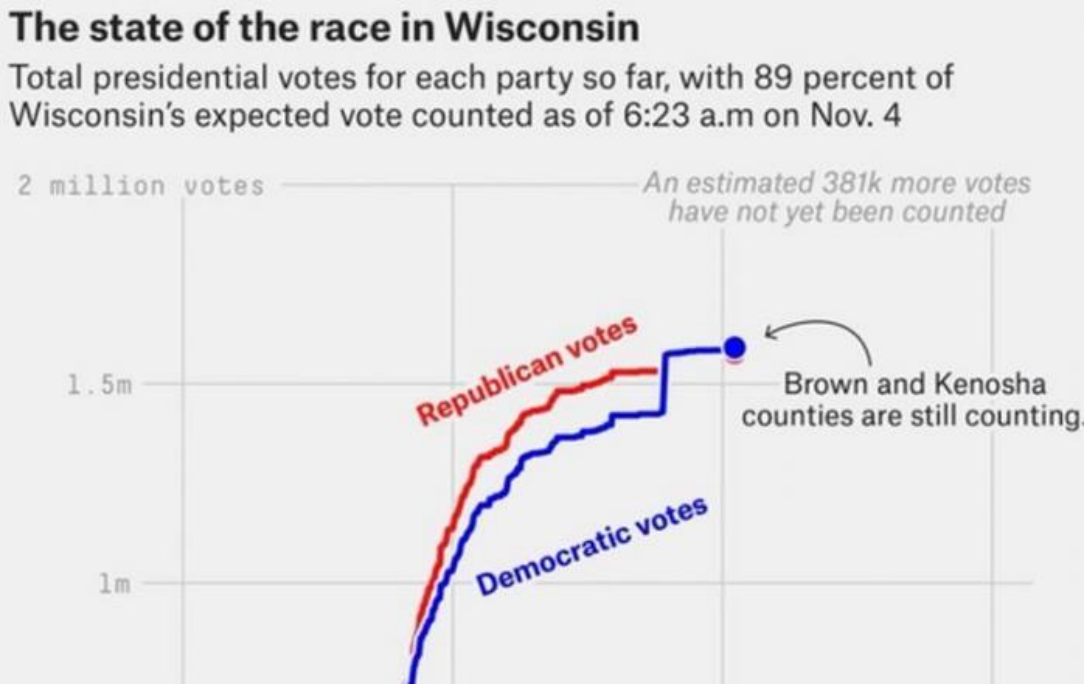
Input – Garbage in, Garbage out

Not Data

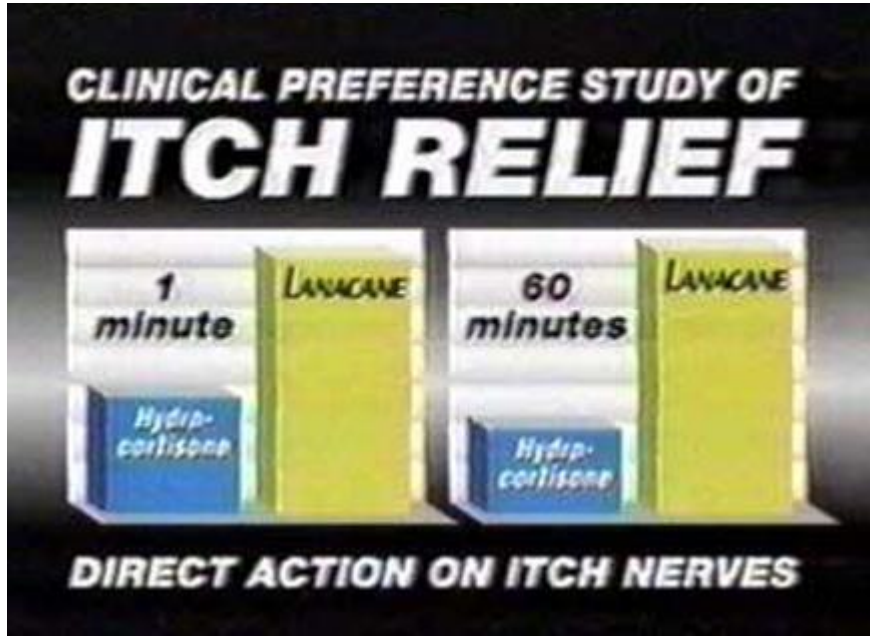
Selective Data

Input – Garbage in, Garbage out

Dubious Data



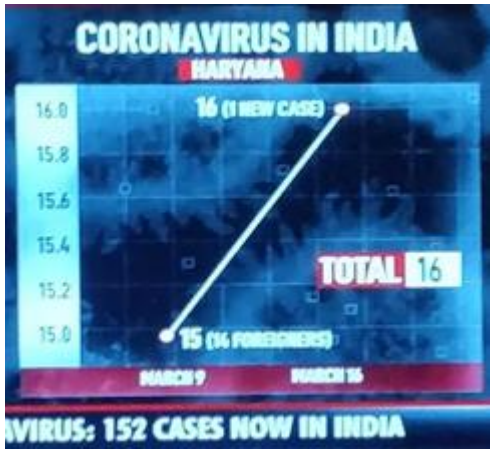
Non Sequitur



(pretends to be a visualization)

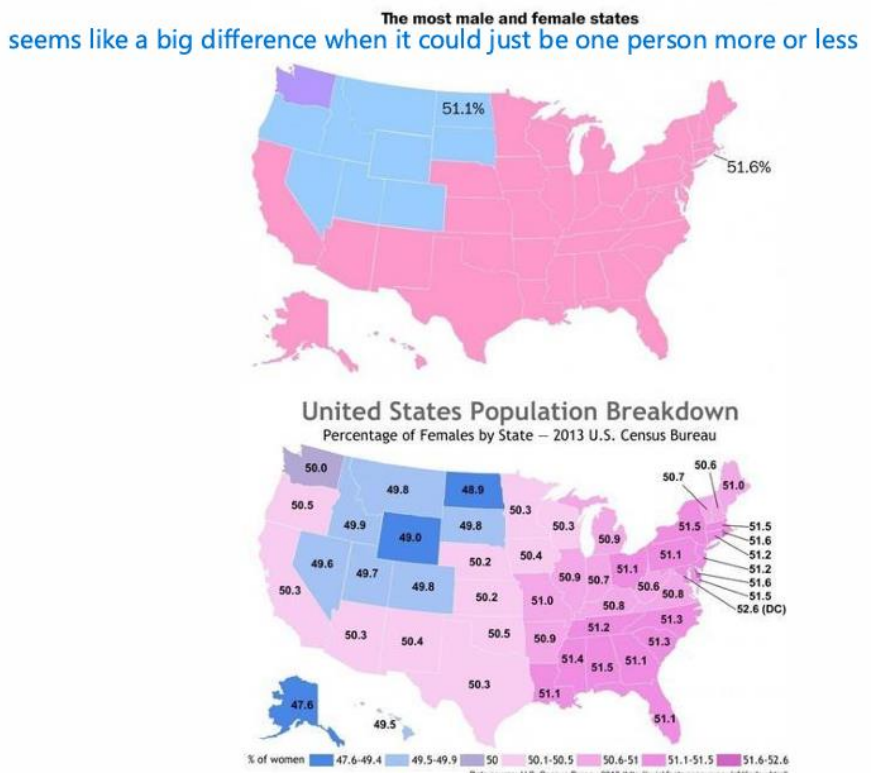
Input – Garbage in, Garbage out

Too Few Data Points

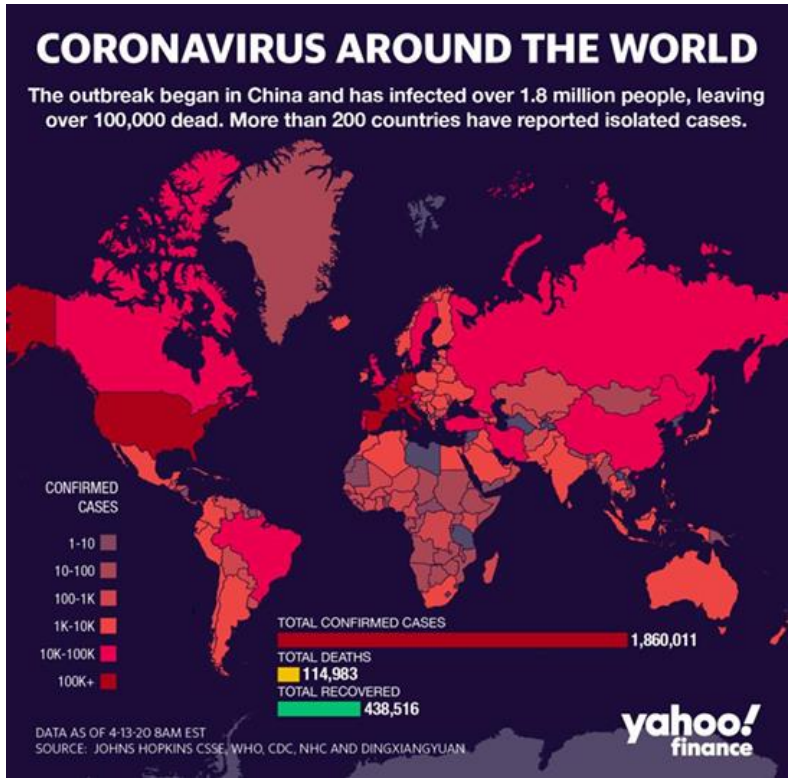


Input – Lie with Statistics

Discretized Continuous Variable

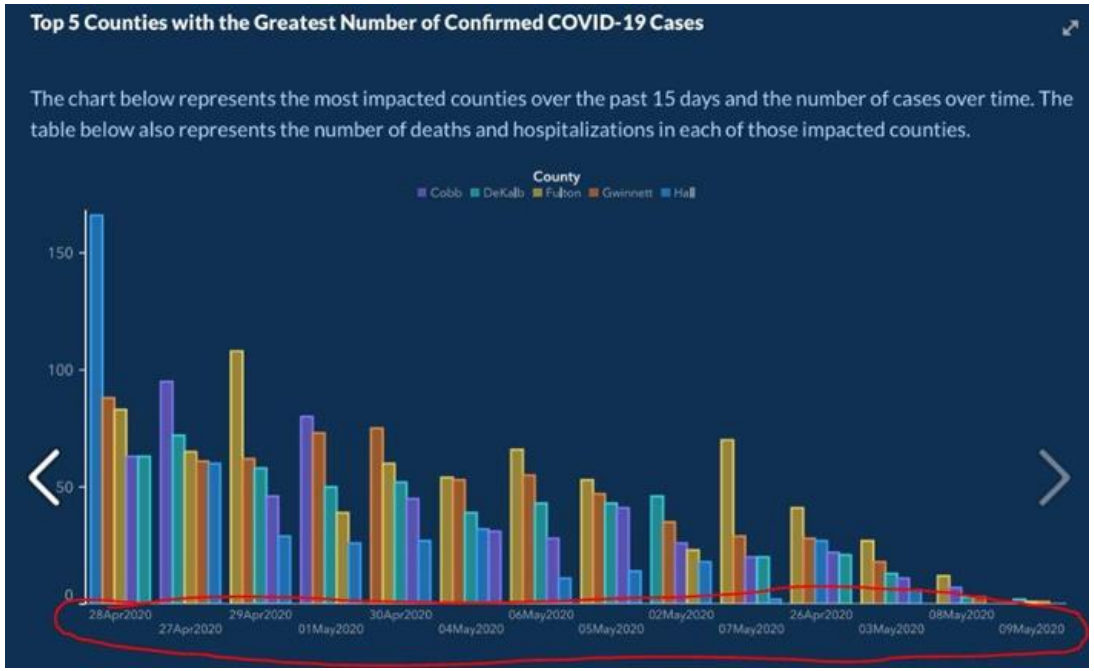


Missing Normalization

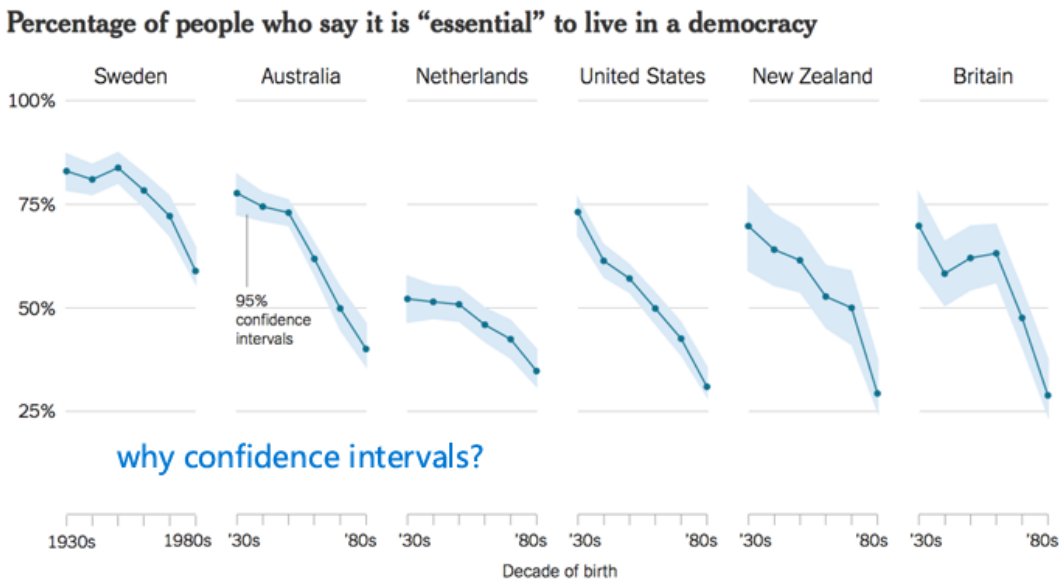


Input – Lie with Statistics

Inappropriate Item Order



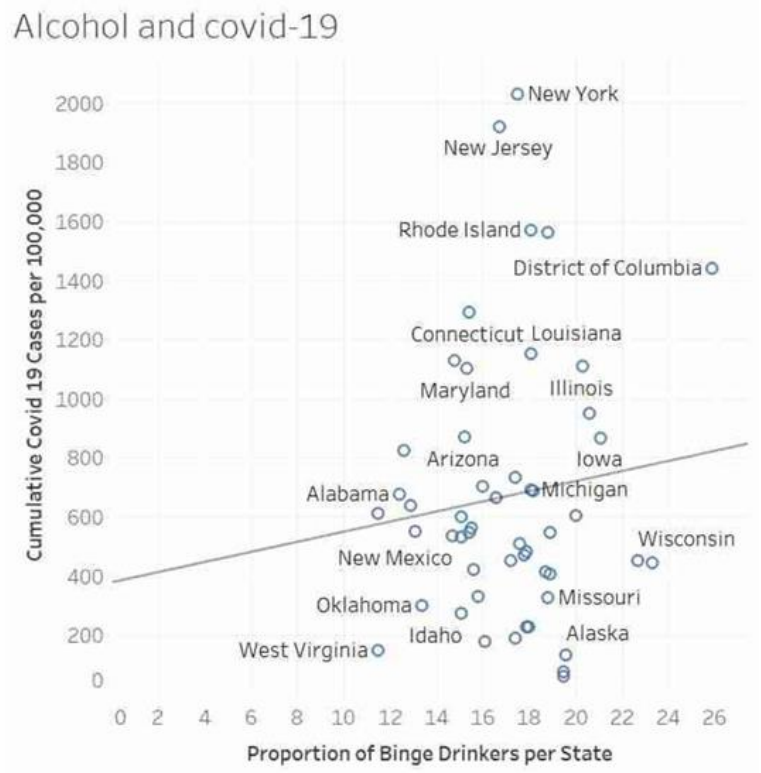
Inappropriate Metric



Input – Lie with Statistics

Questionable Prediction

Trend Line on Random Data



Input – Lie with Statistics

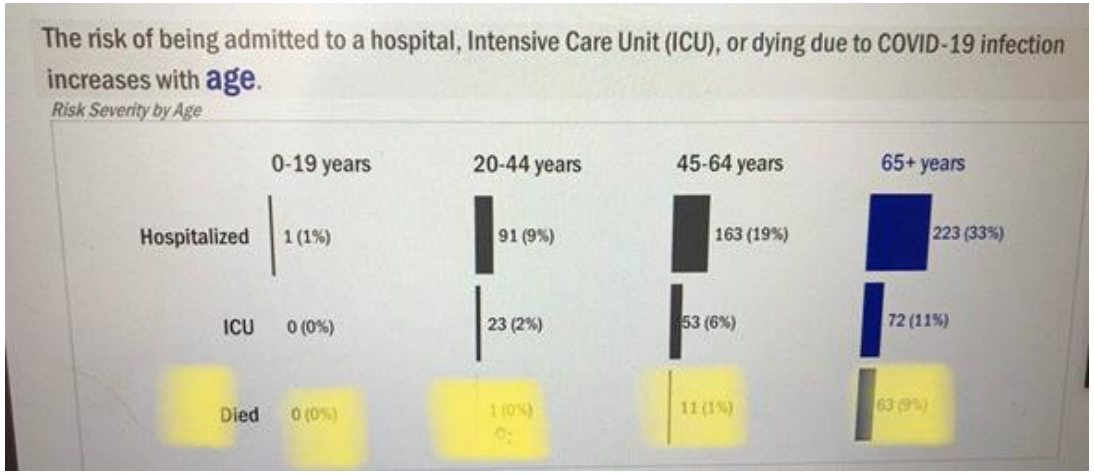
Inappropriate Use of Accumulation

Inappropriate Aggregation Granularity

Percentage of population using "Irish Gaelic"
-Too much aggregation in Northern Ireland

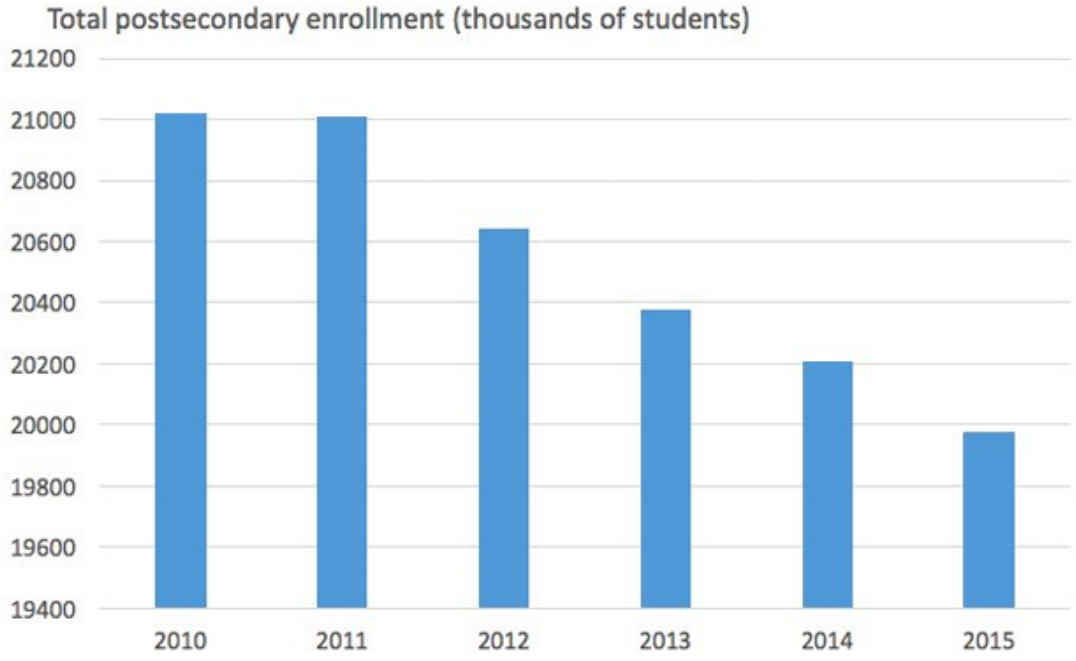
Input – Lie with Statistics

Two-Way Normalization

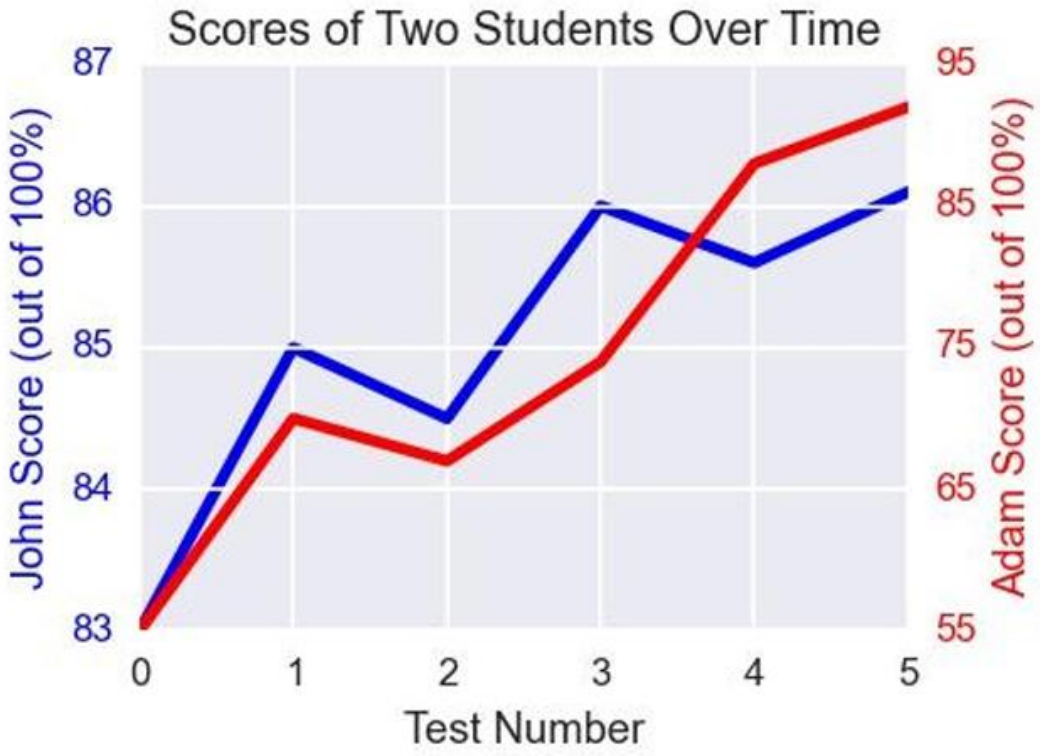


Visualization Design – Choice of Axis

Truncated Axis

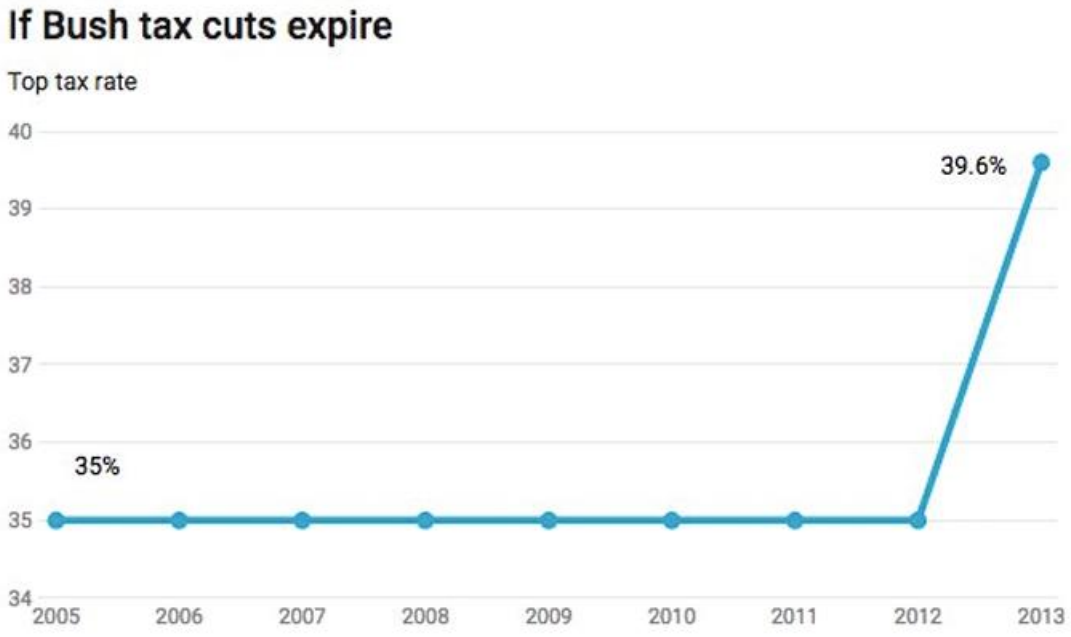


Dual Axis

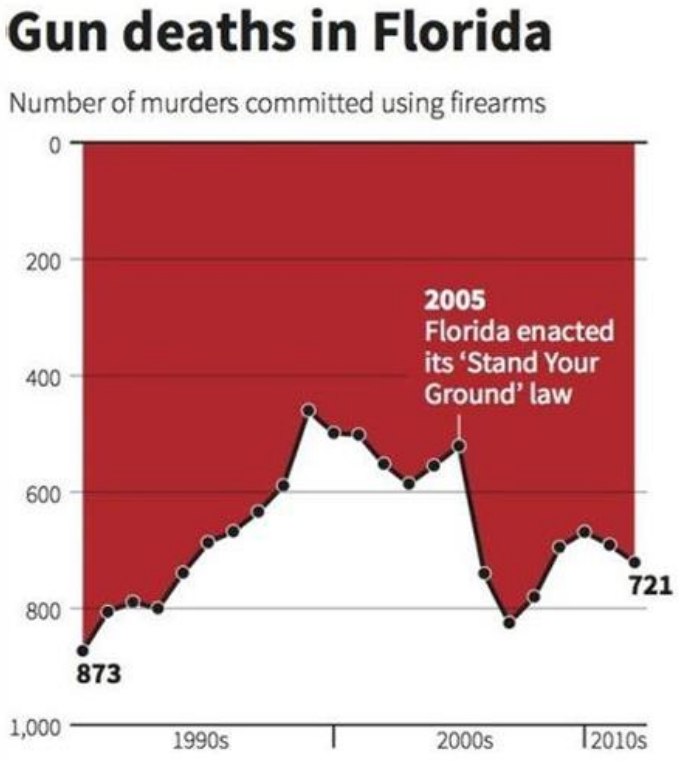


Visualization Design – Choice of Axis

Inappropriate Axis Range

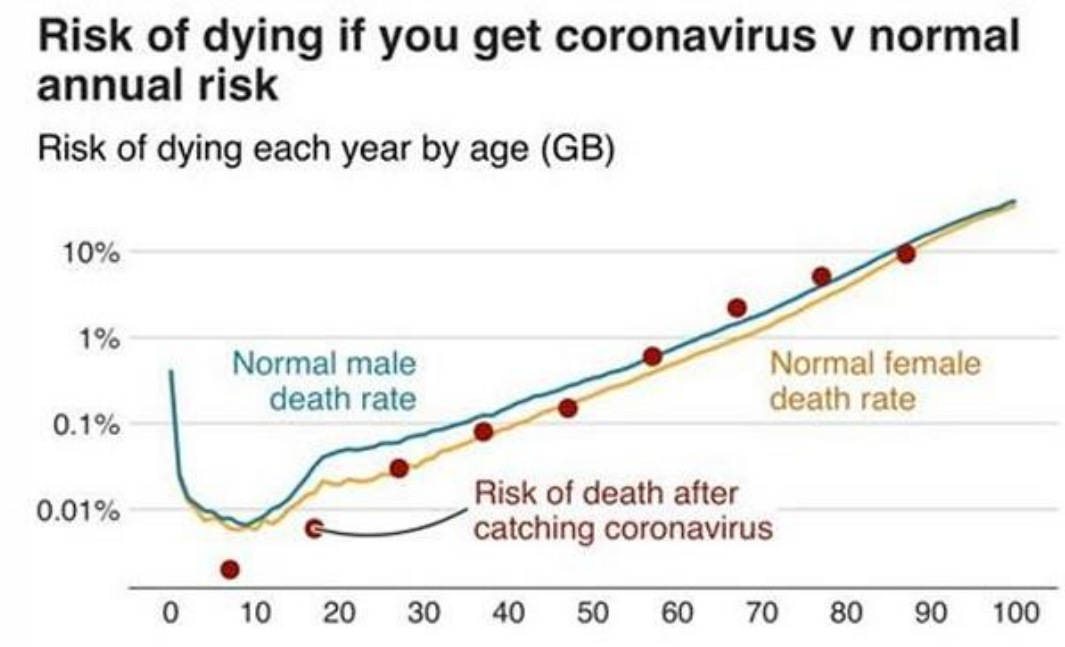


Inverted Axis

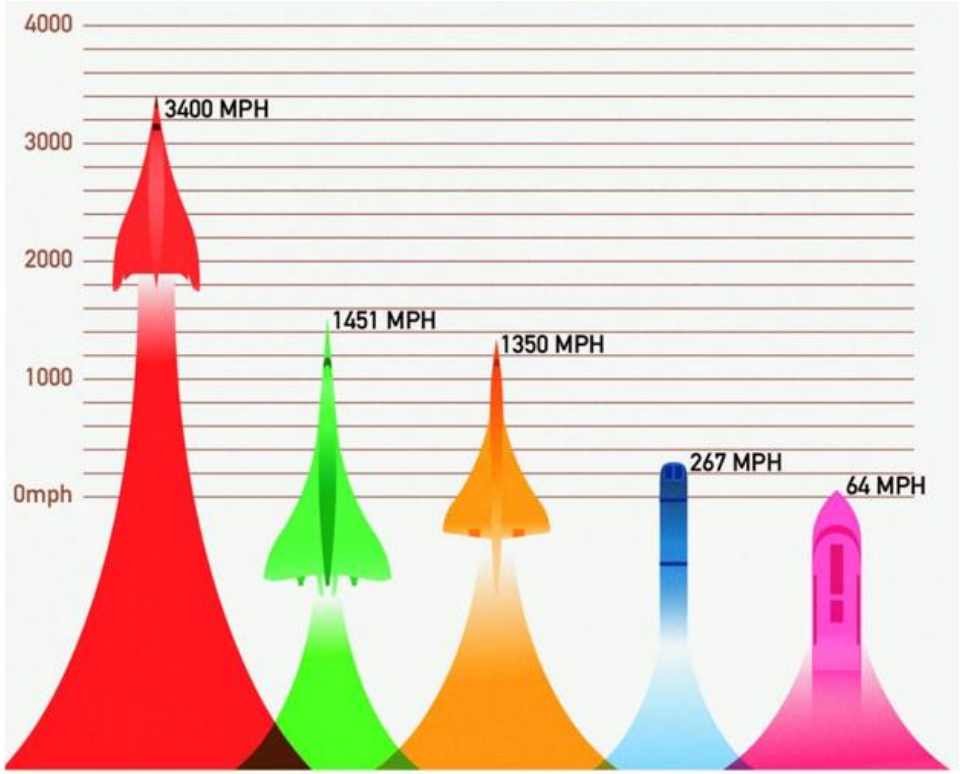


Visualization Design – Choice of Axis

Log Scale

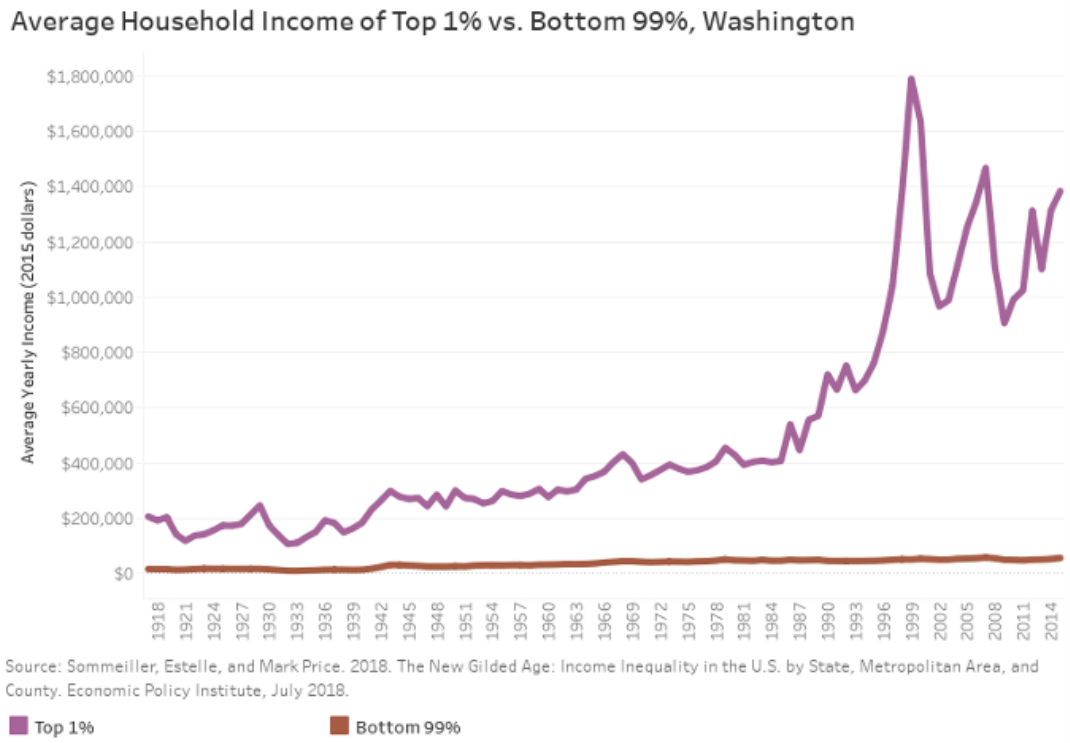


Extended Axis

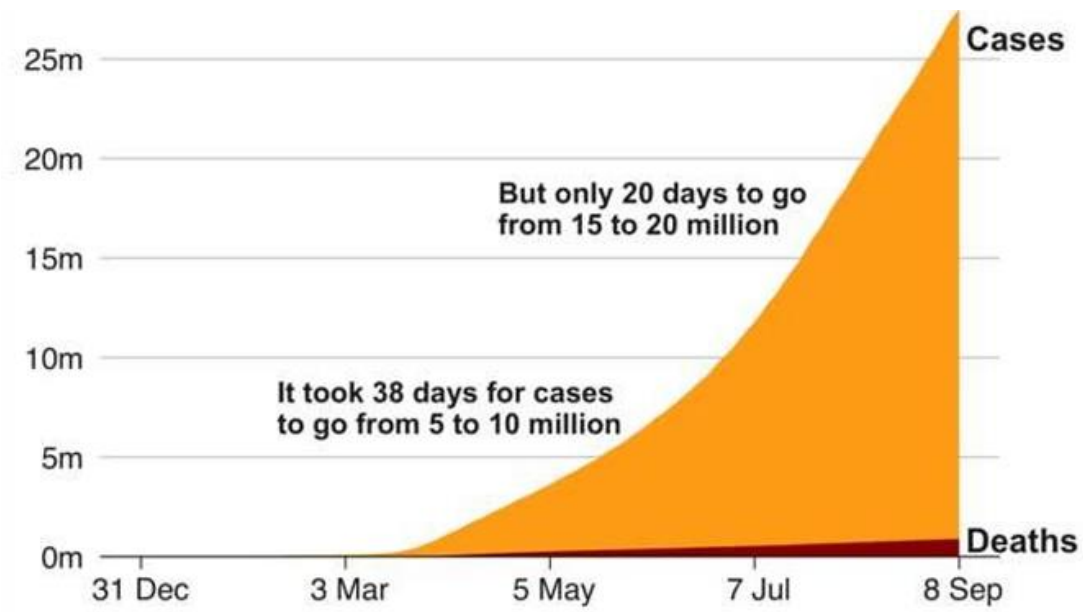


Visualization Design – Choice of Axis

Data of Different Magnitudes



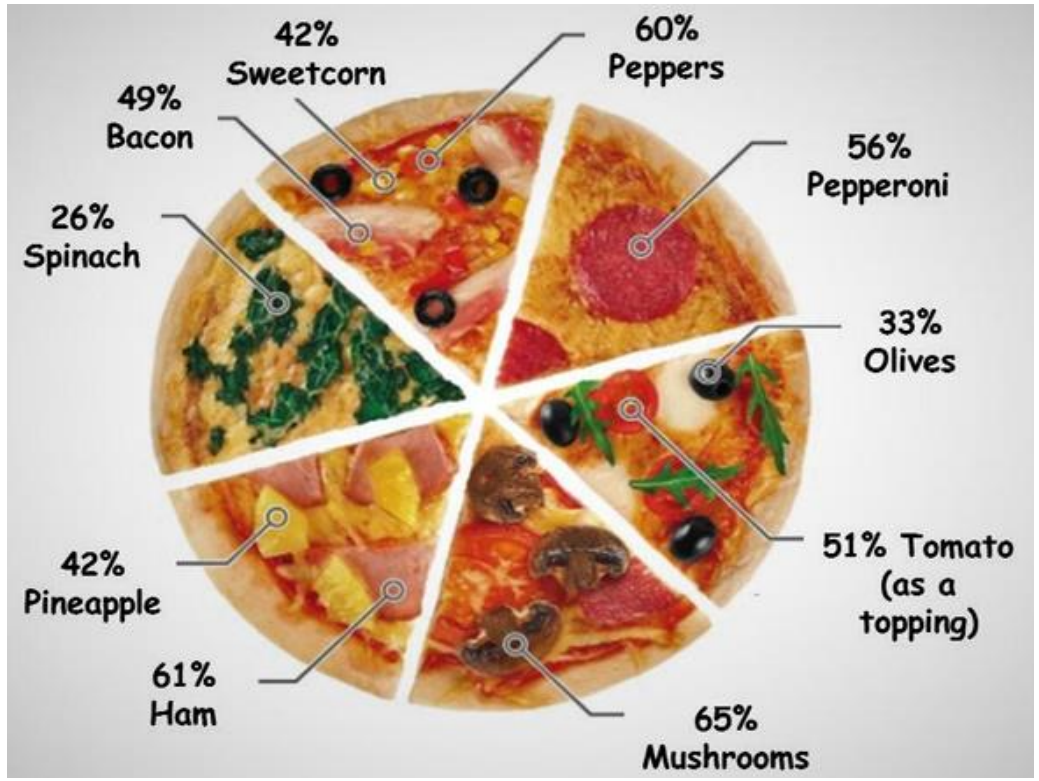
Linear Scale on Exponential Data



Visualization Design – Choice of Chart

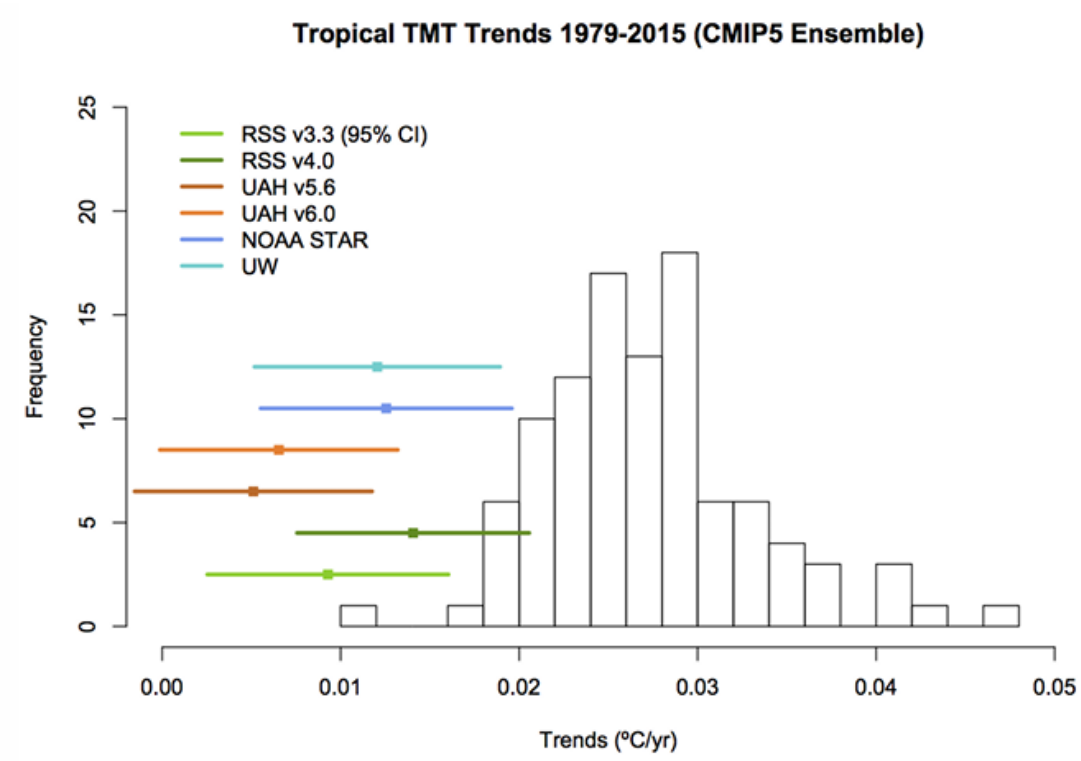
Inappropriate Use of Line Chart

Inappropriate Use of Pie Chart

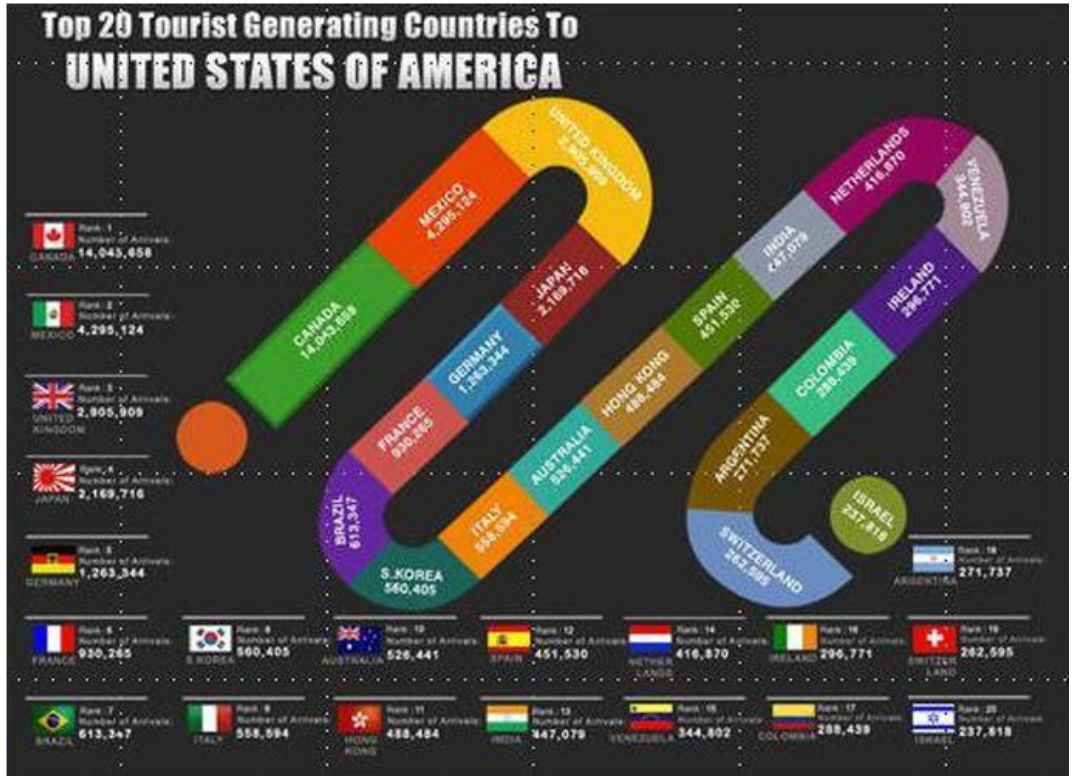


Visualization Design – Choice of Chart

Confusing Chart Type

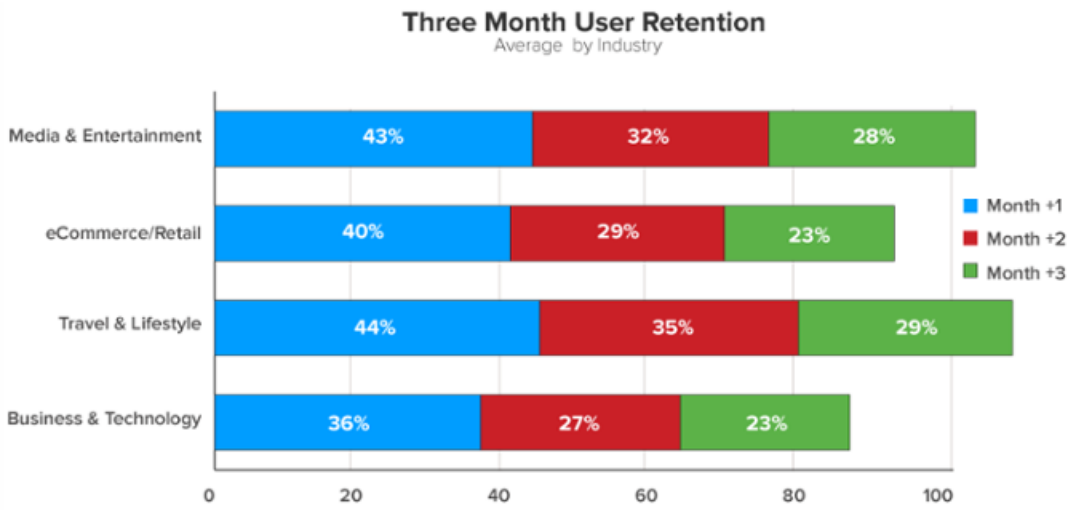


Misusing Circular Layout

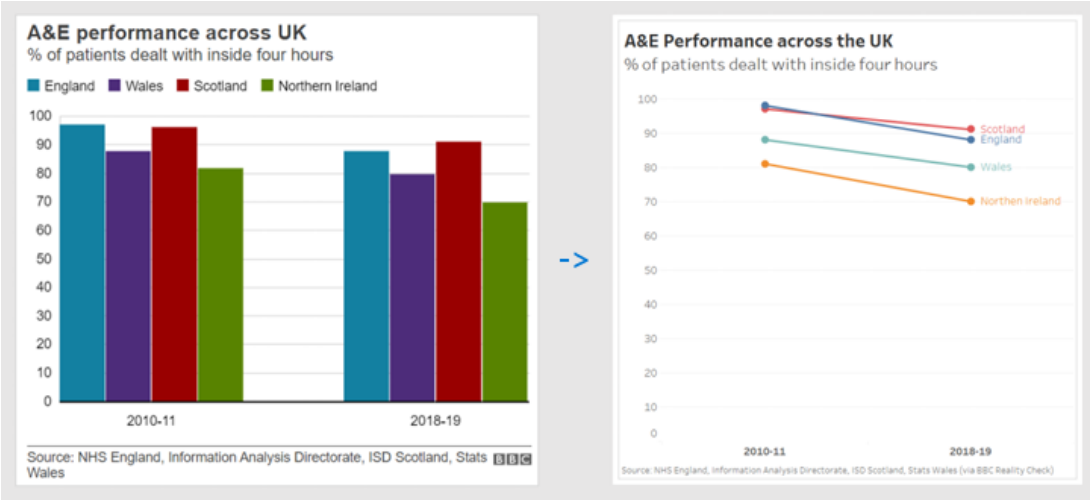


Visualization Design – Choice of Chart

Inappropriate Use of Stacked Chart

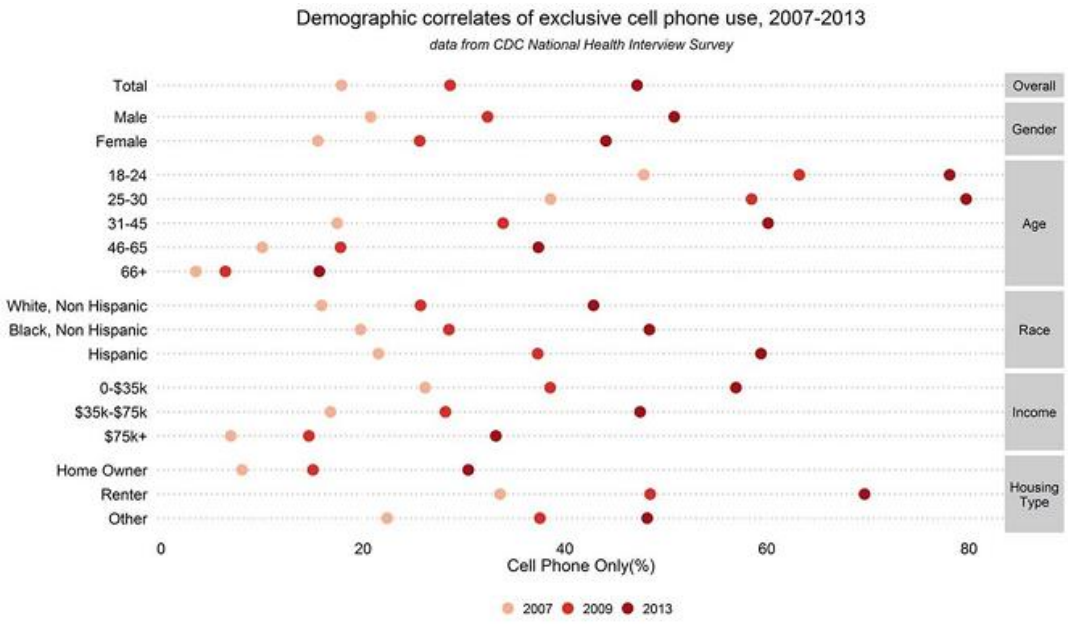


Inappropriate Use of Bar Chart



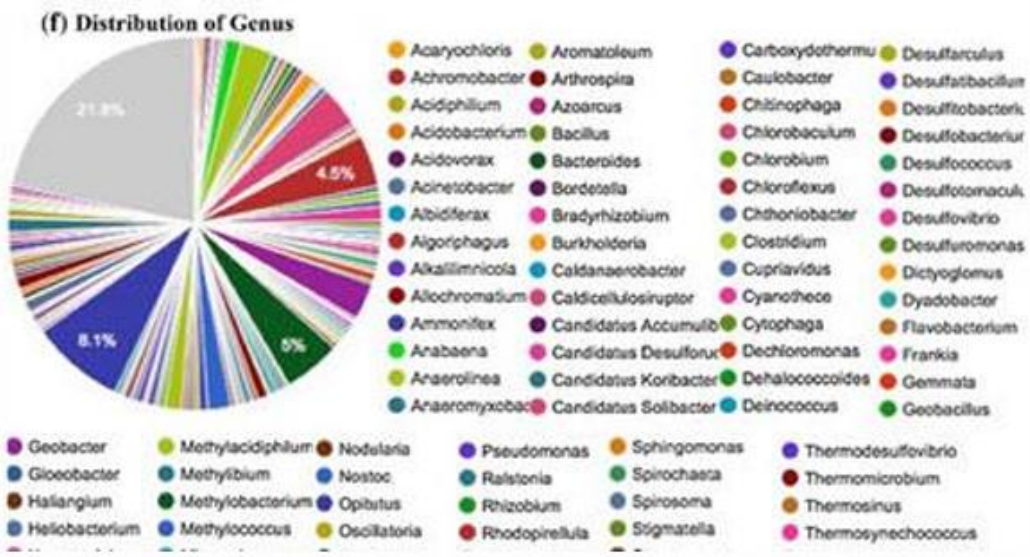
Visualization Design – Choice of Chart

Inappropriate Use of Scatterplot

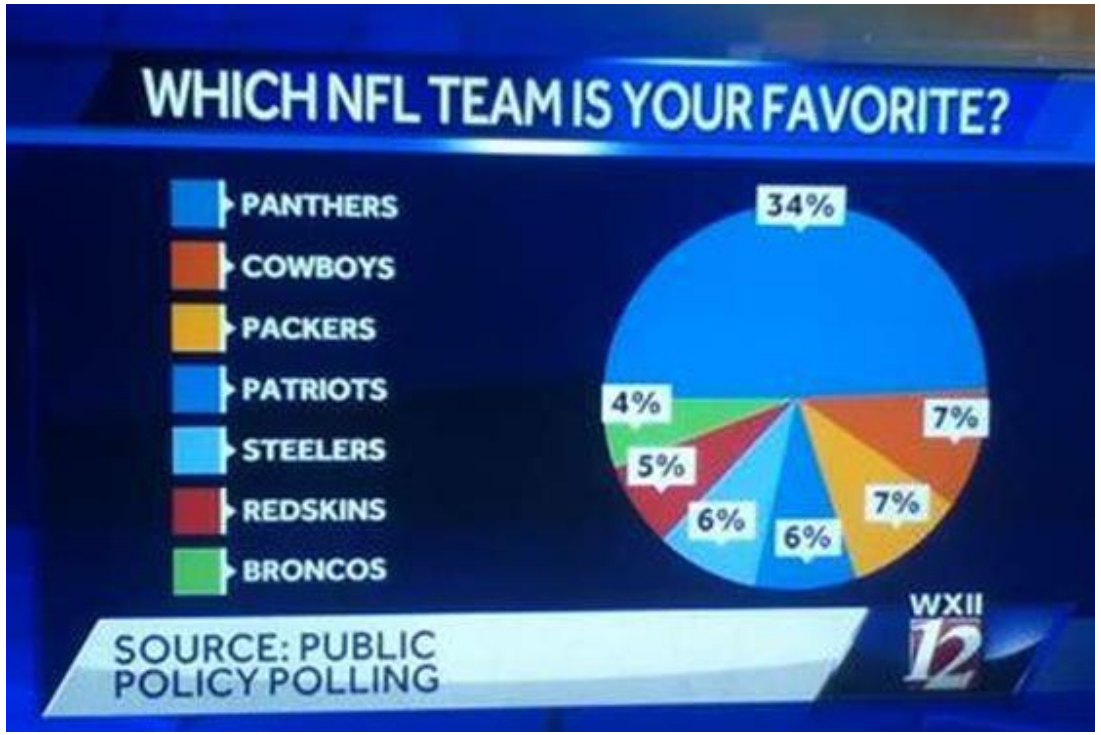


Visualization Design – Color Mess

Overusing Colors



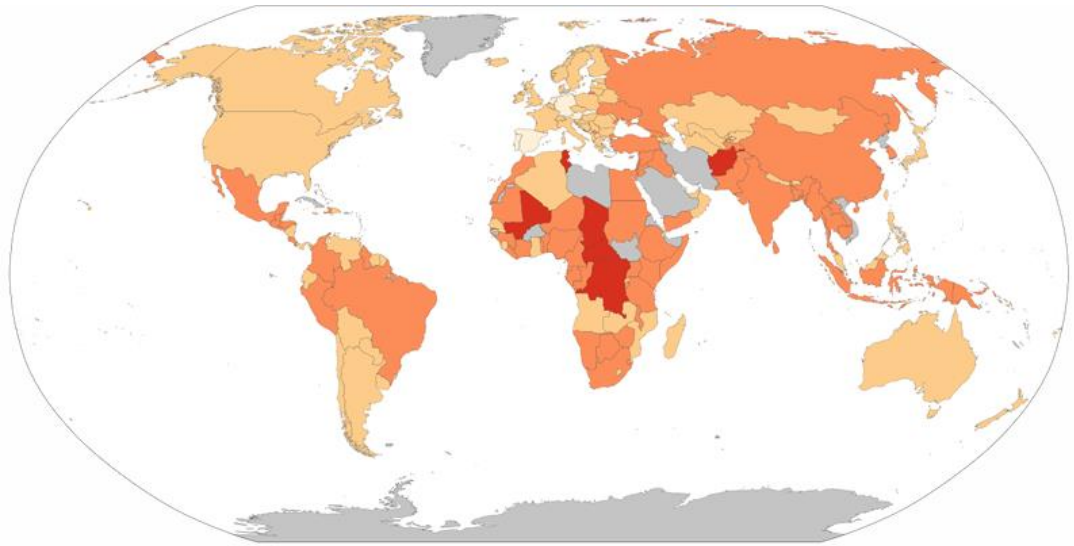
Indistinguishable Colors



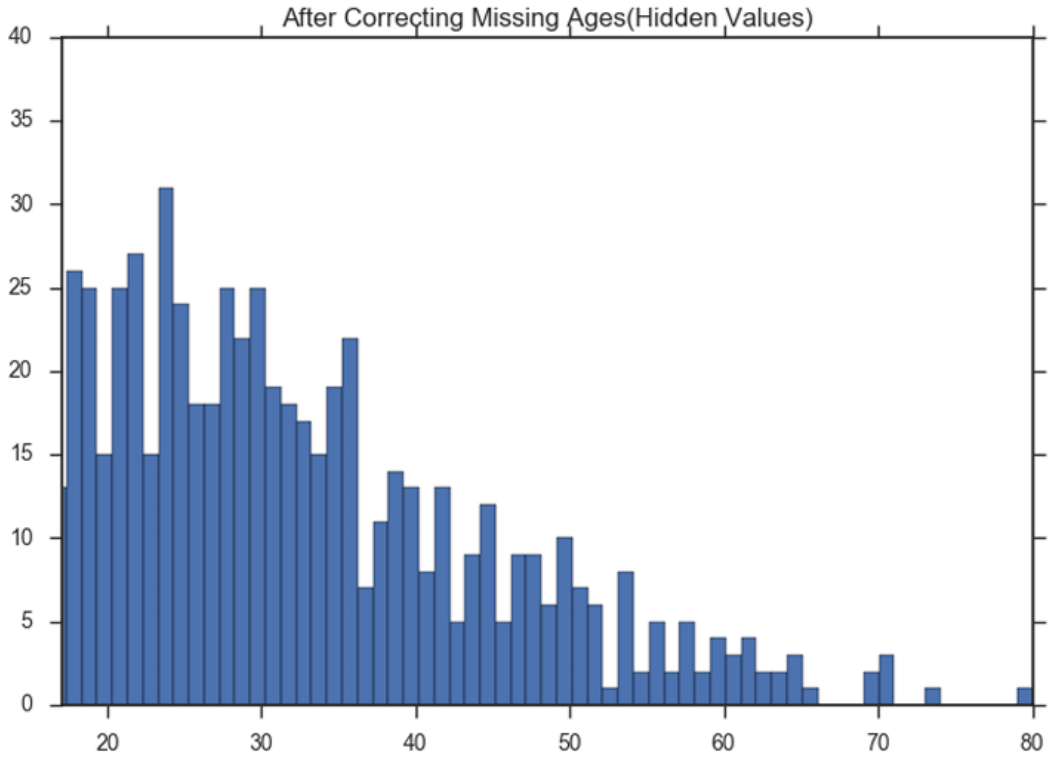
Also think about color-blindness!

Plotting – Incomplete Chart

Missing Title

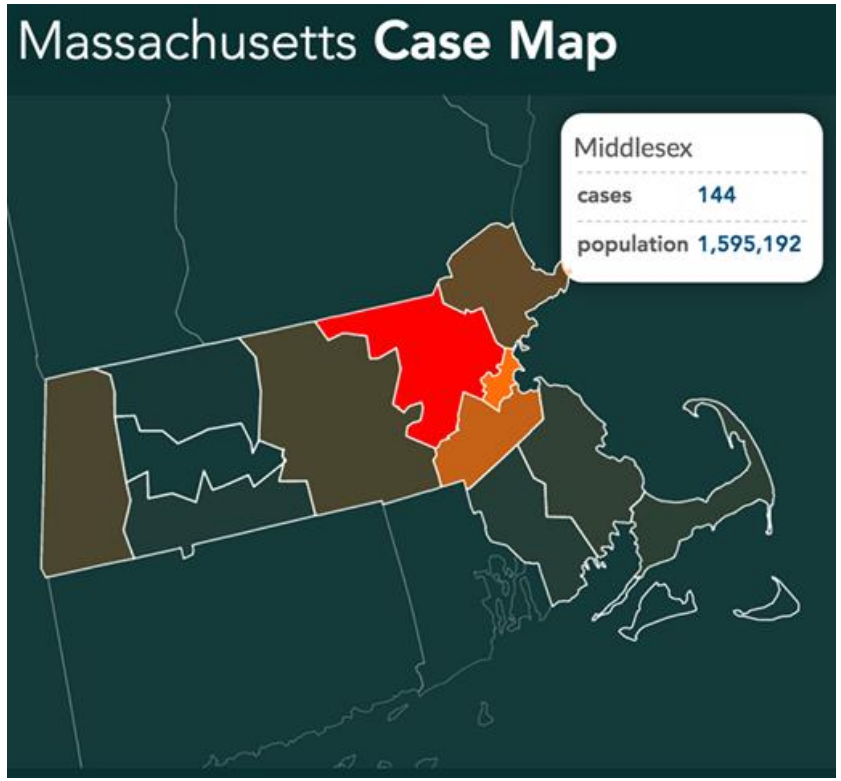


Missing Axis Title

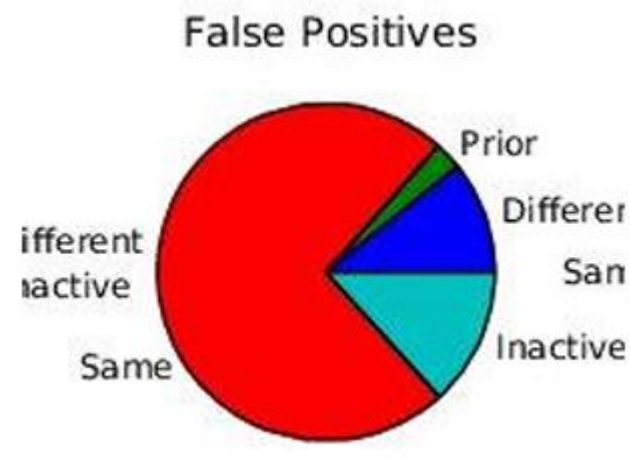


Plotting – Incomplete Chart

Missing Legend

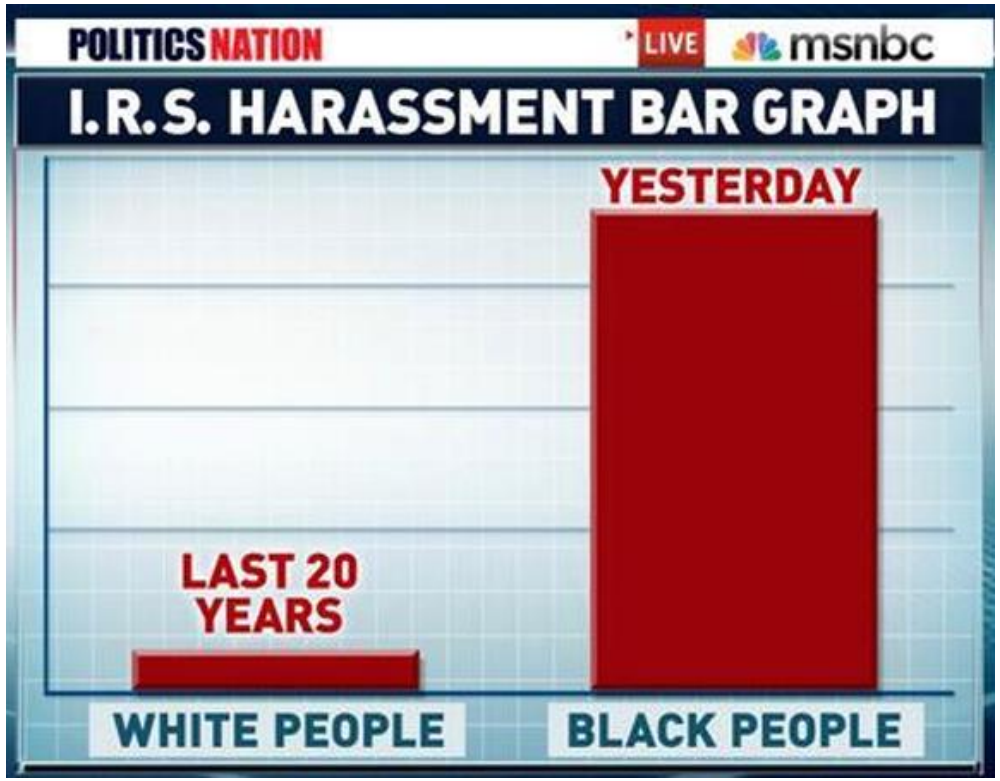


Missing Value Labels

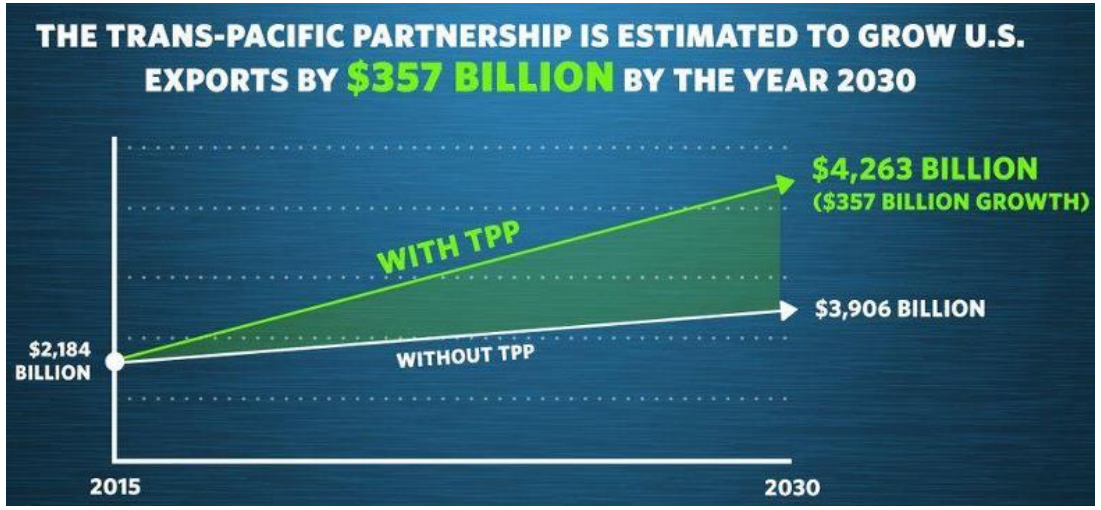


Plotting – Incomplete Chart

Missing Axis

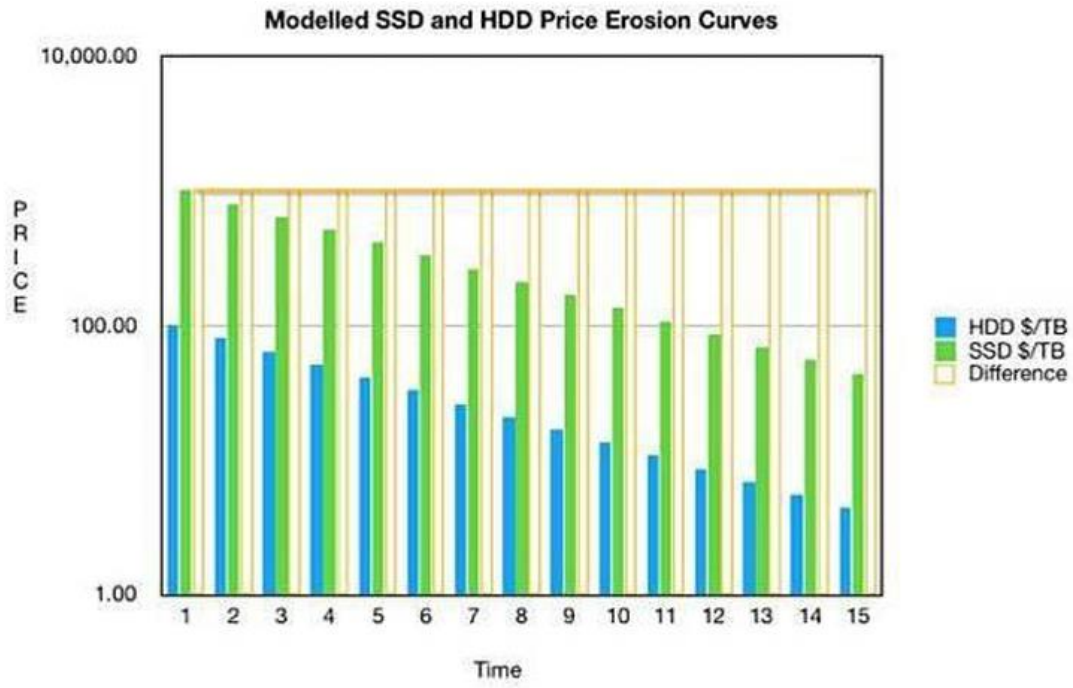


Missing Axis Ticks



Plotting – Incomplete Chart

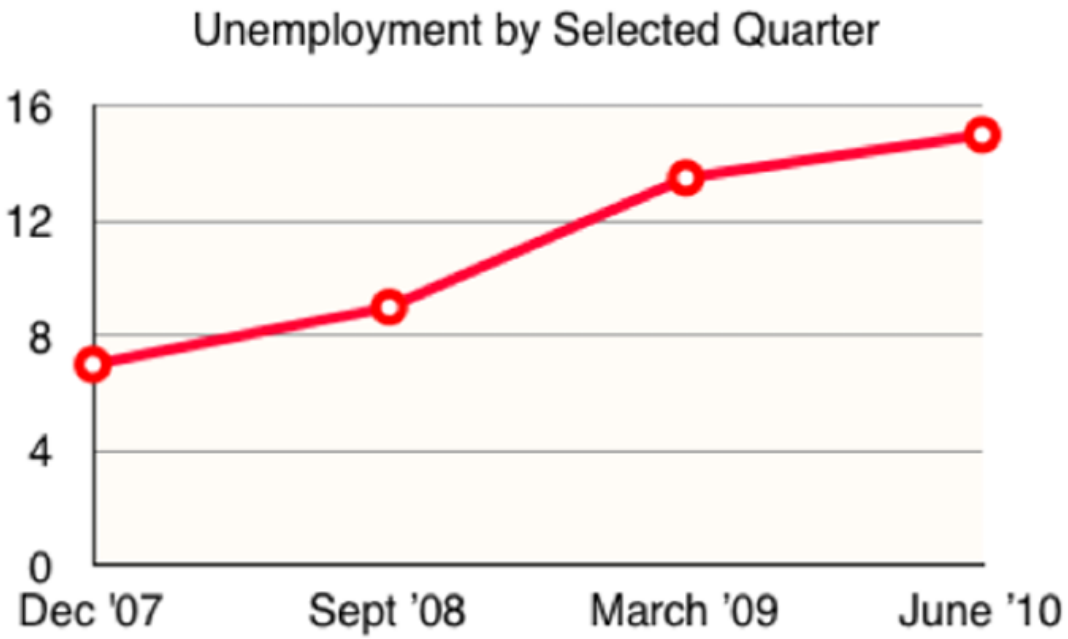
Missing Units



Plotting – Inconsistency

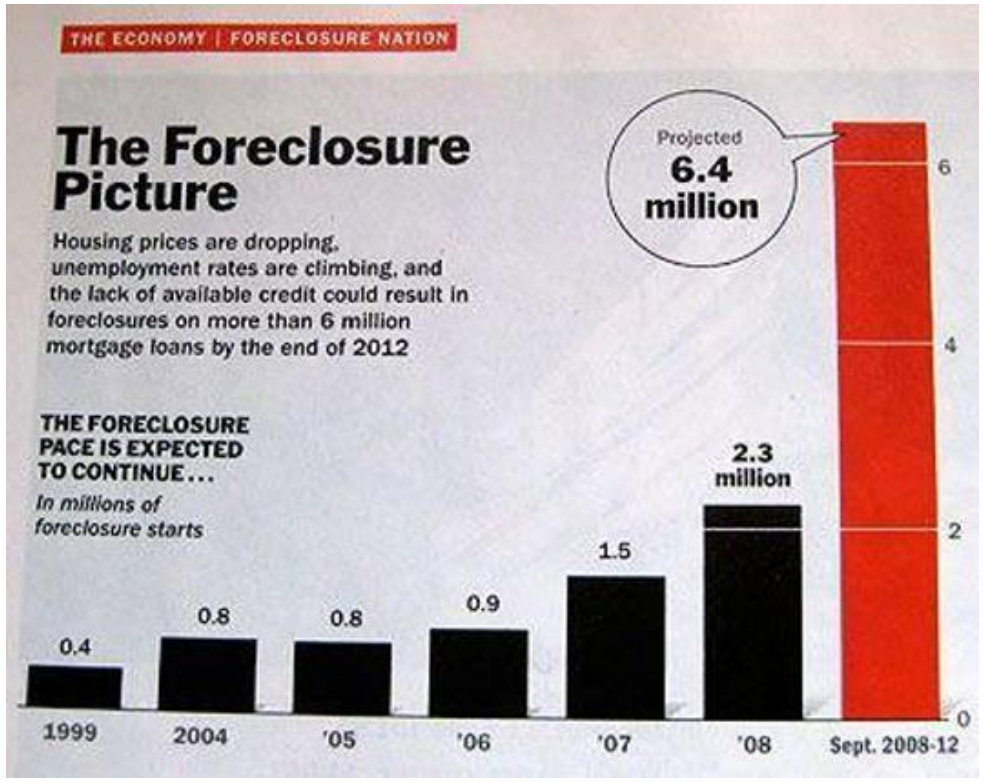
Misrepresentation

Inconsistent Tick Intervals

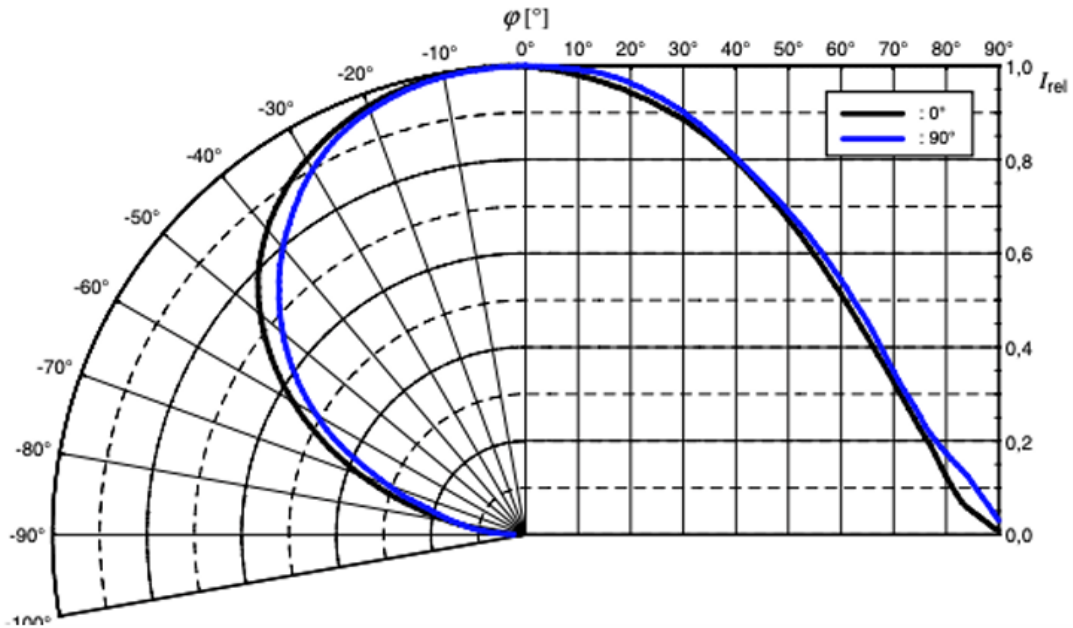


Plotting – Inconsistency

Inconsistent Binning Size

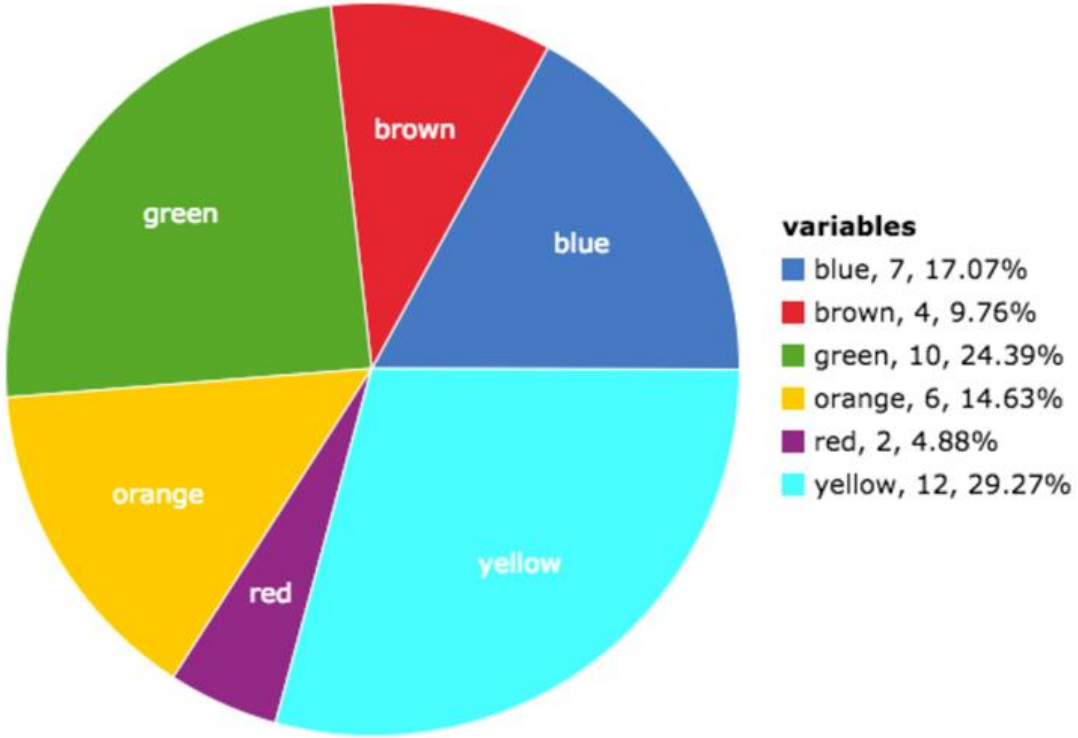


Changing Scale

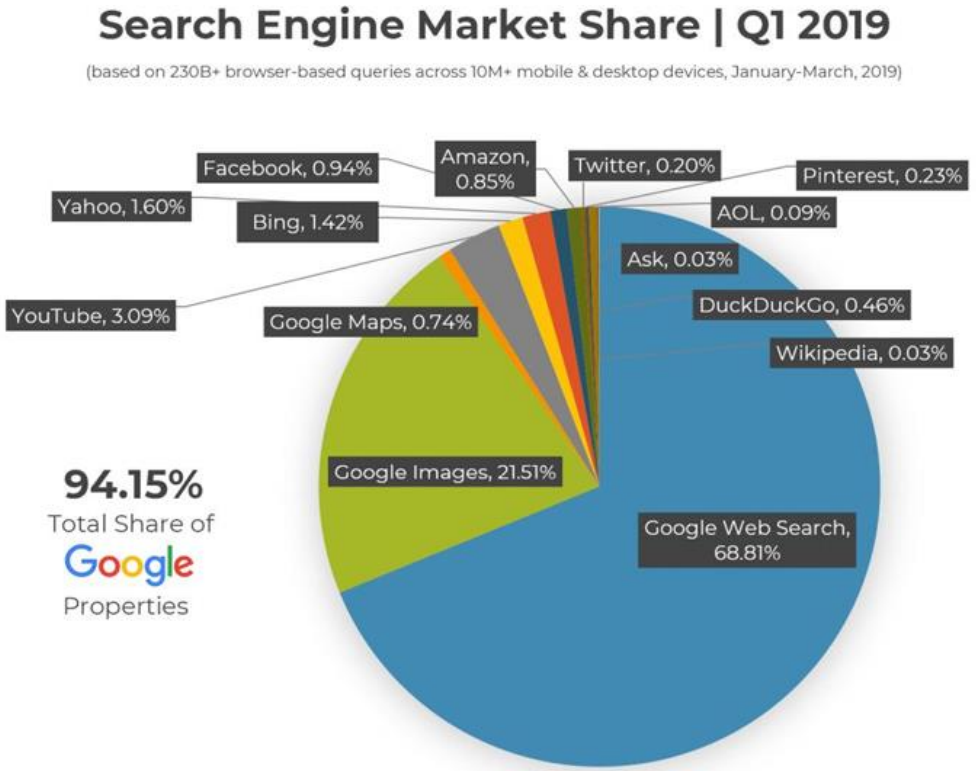


Plotting – Inconsistency

Violating Color Convention

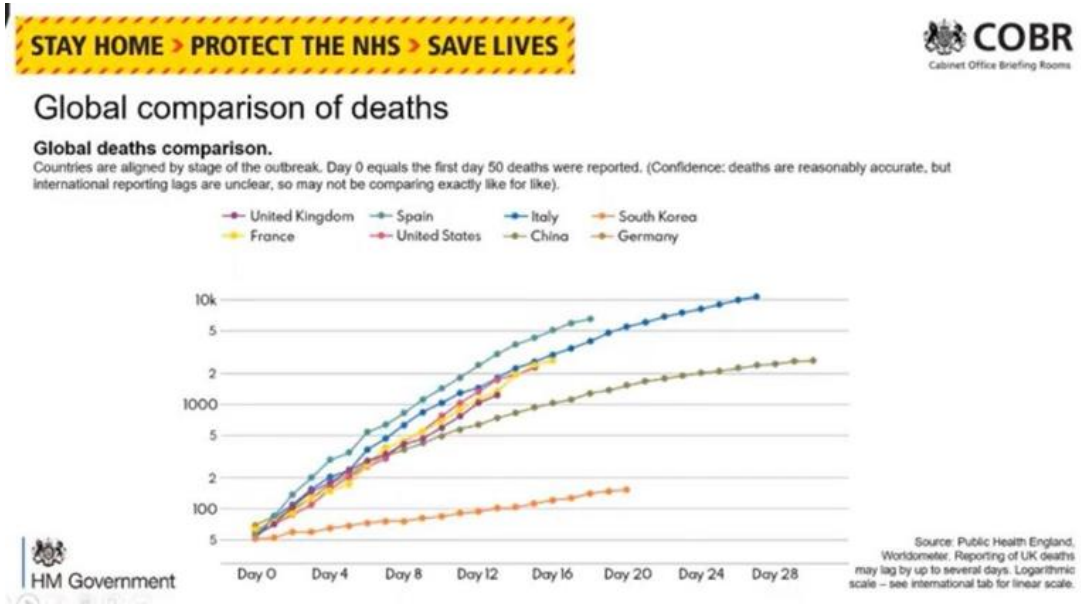


Inconsistent Grouping

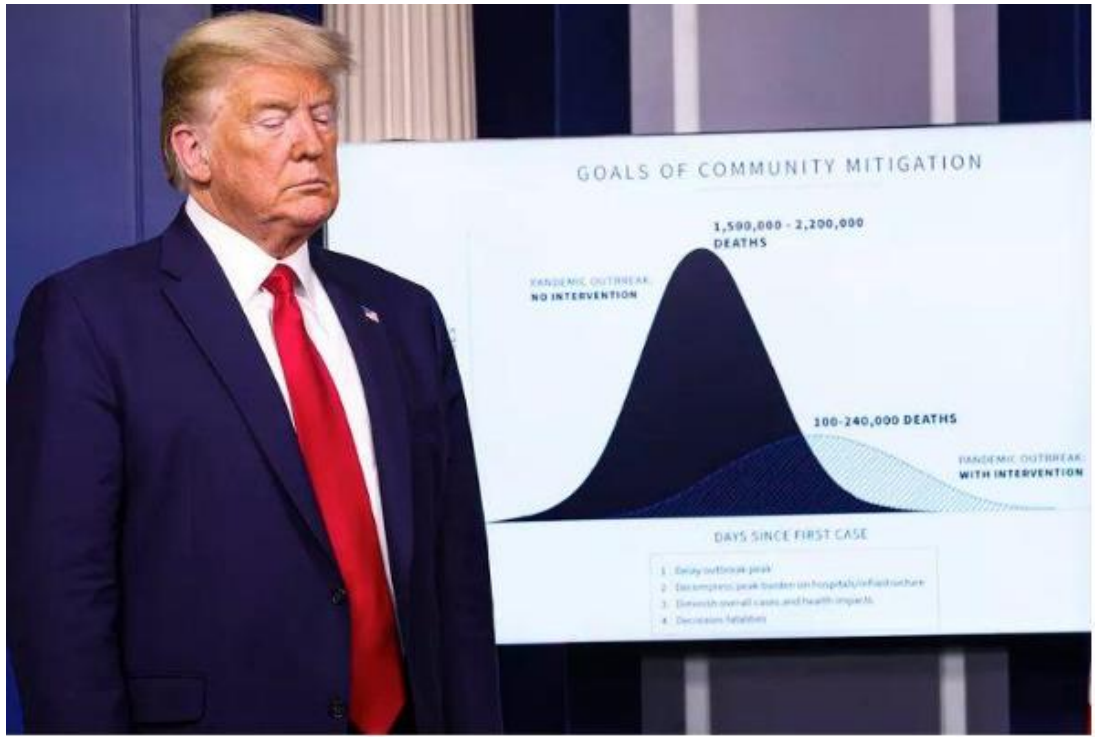


Plotting – Inconsistency

Inconsistent Tick Labels

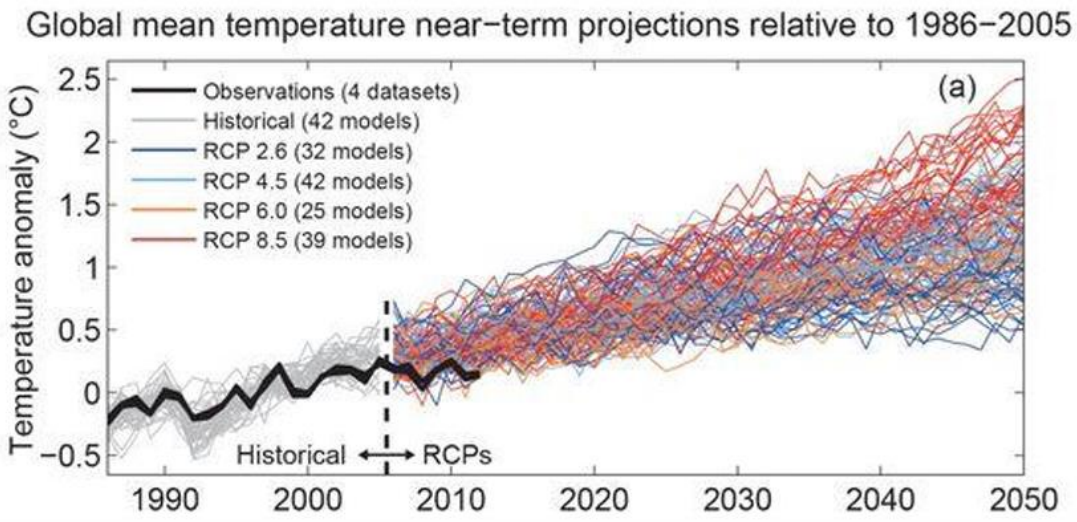


Inconsistent Value Labels

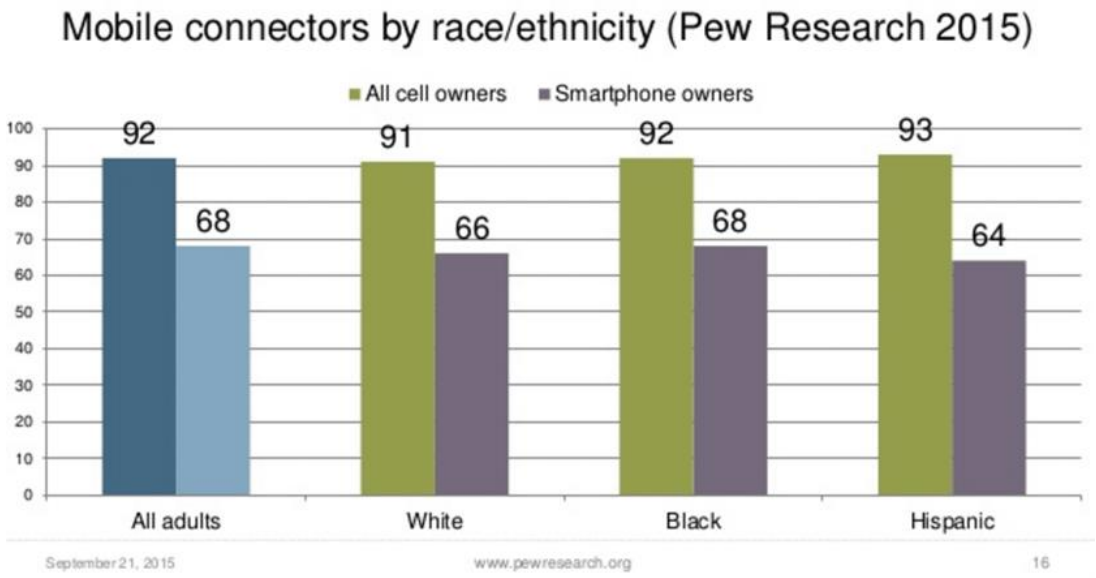


Plotting – Chaotic Canvas

Cluttering

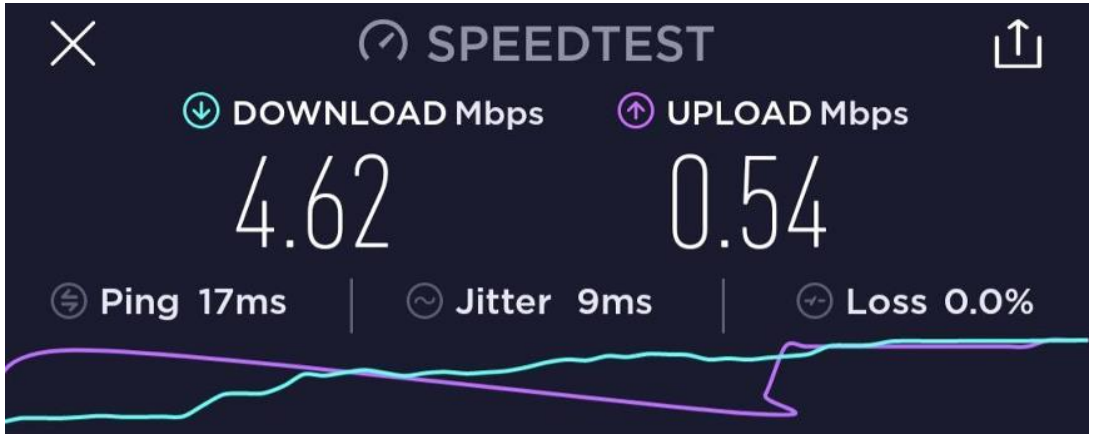


Confusing Legend

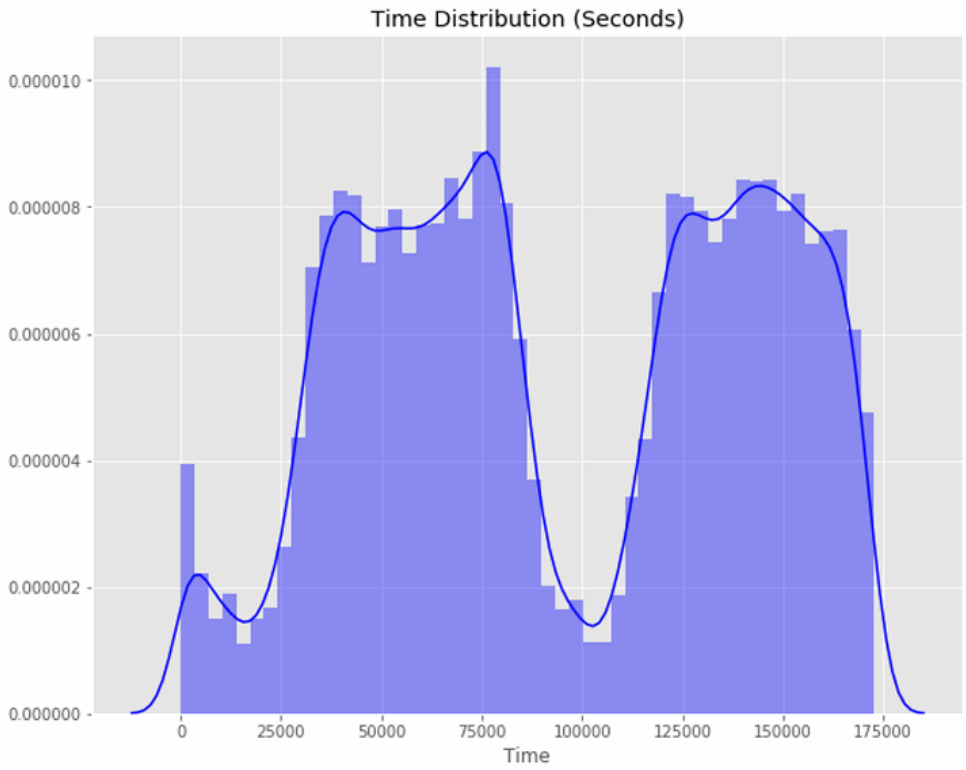


Plotting – Chaotic Canvas

Plotting Error

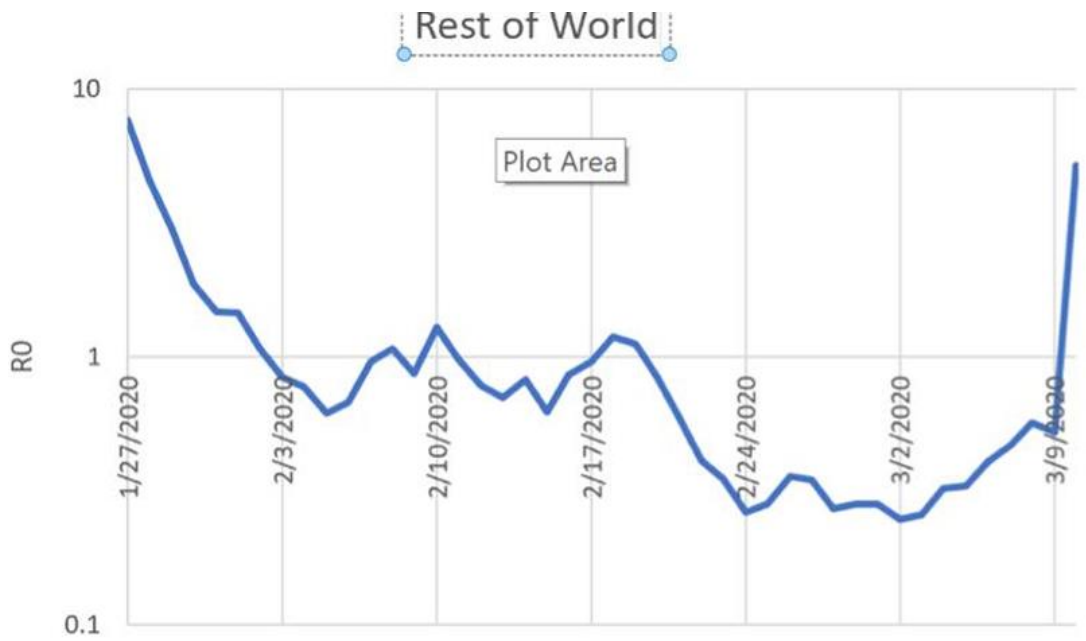


Missing Abbreviation

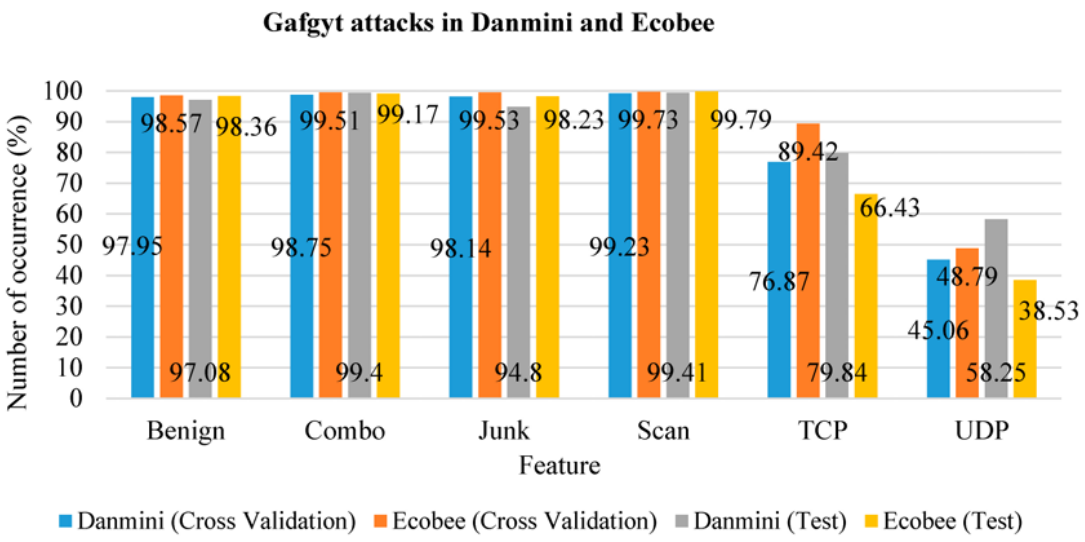


Plotting – Chaotic Canvas

Plotting Out of Chart

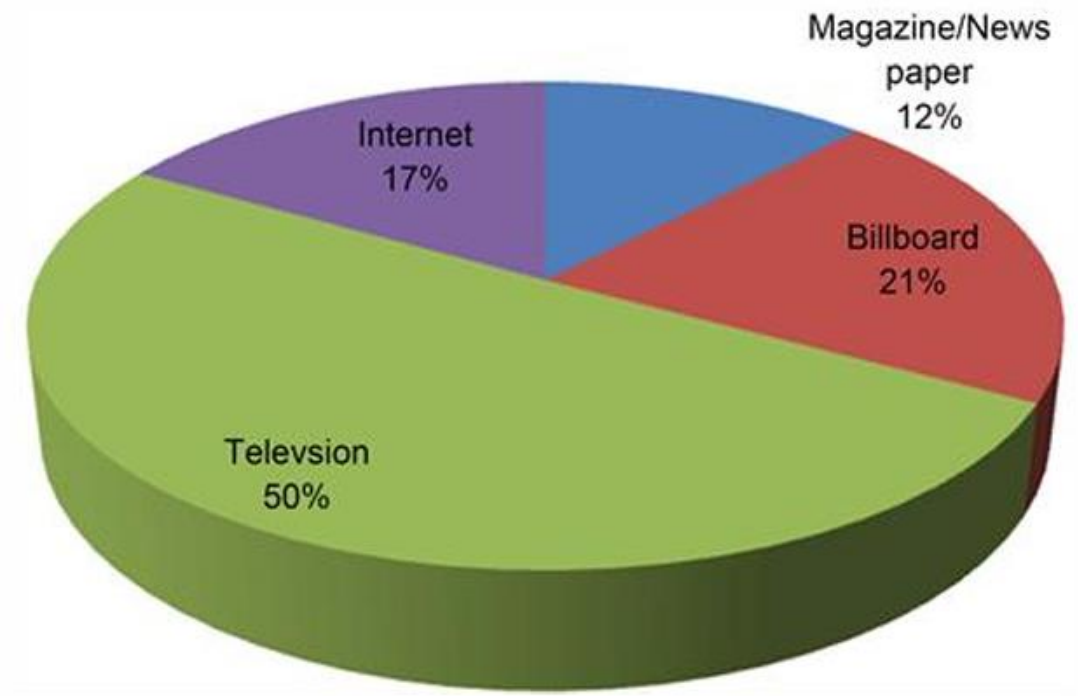


Misalignment

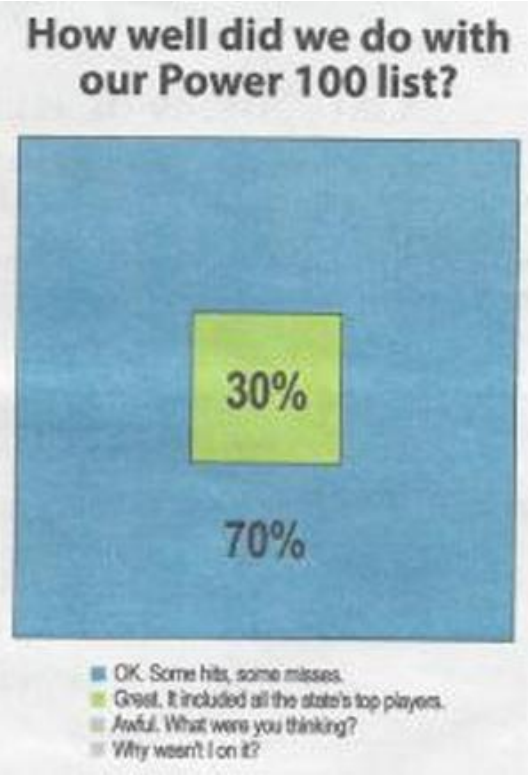


Perception – Visual Illusion

3D

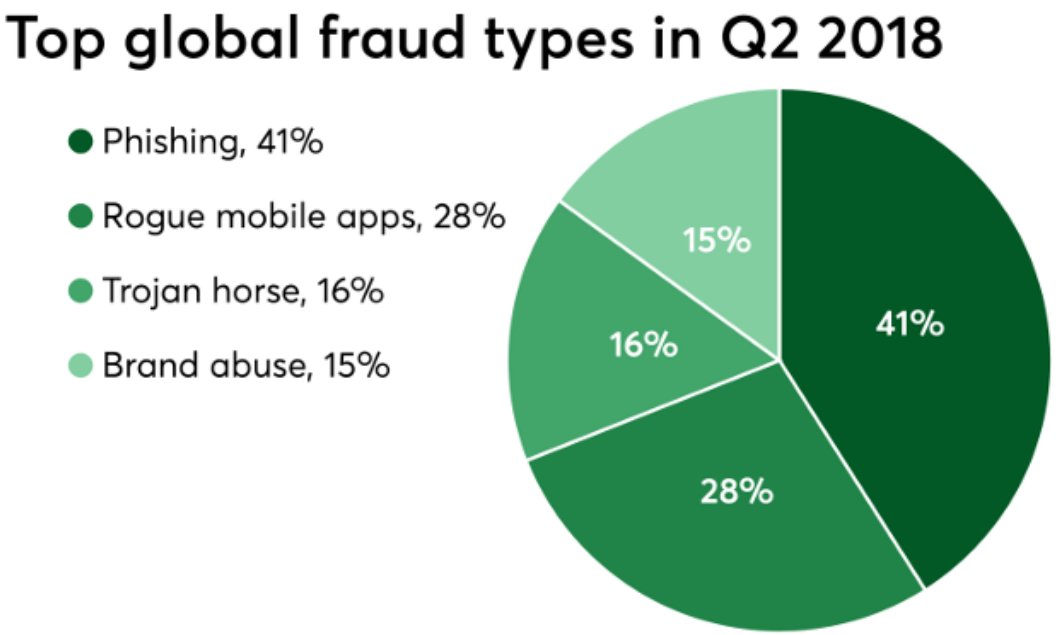


Area Encoding

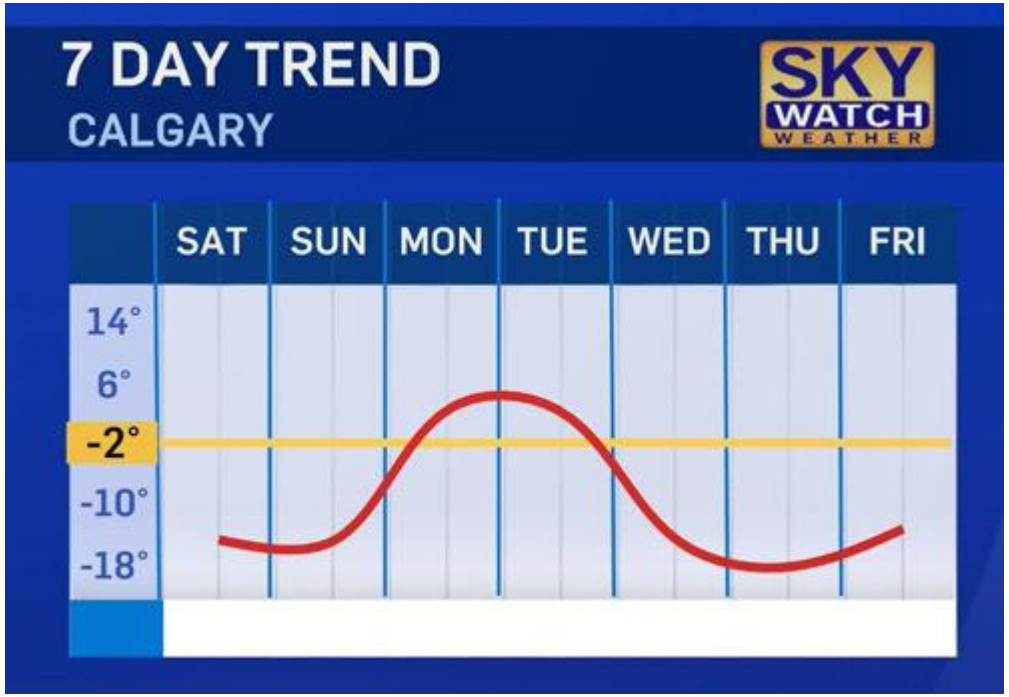


Perception – Visual Illusion

Ineffective Color Scheme



Inappropriate Use of Smoothing



Perception – Visual Illusion

Distractive Value Labels

FIGURE 2B:

Breakdown of complaints in other financial services category

Fraud or scam is the most complained about topic in this category

Fraud or scam 46%

Customer service or customer relations 21%

Excessive fees 7%

^ perceived as larger %

Unexpected or other fees 7%

Advertising and marketing 6%

Other 5%

Disclosures 5%

Lost or stolen check 2%

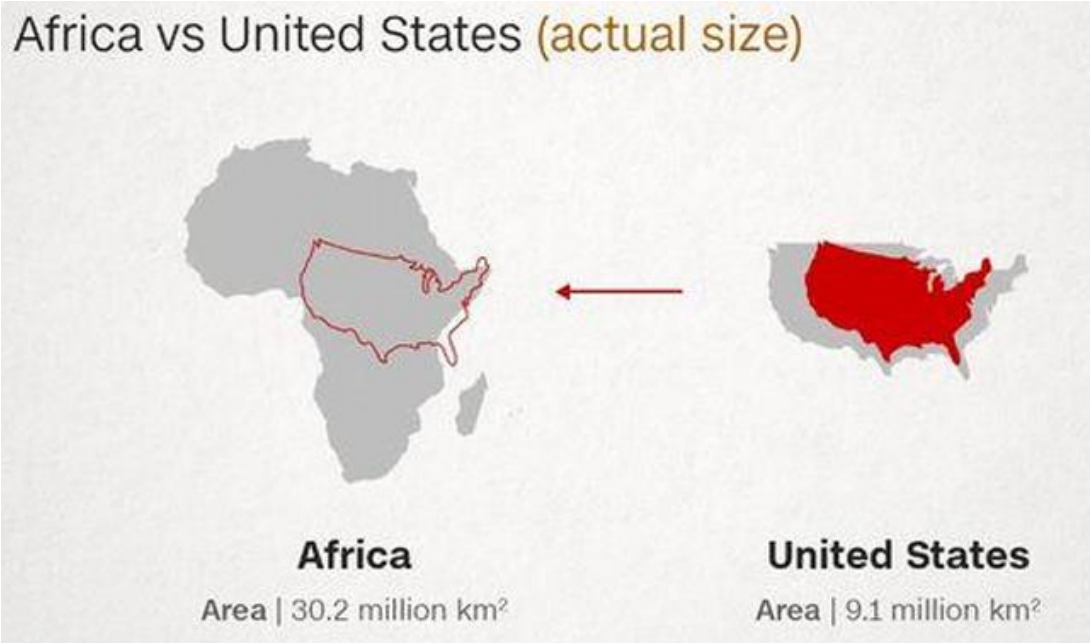
Lost or stolen money order 1%

Incorrect exchange rate 1%

Source: https://data.consumerfinance.gov/dataset/Consumer-Complaints/s6ew-h6mp

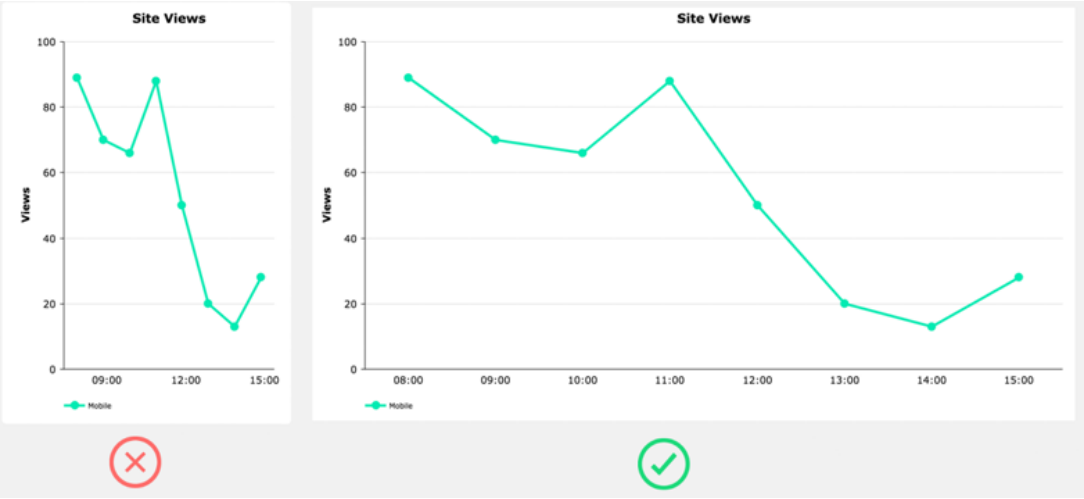
Numbers add up to more than 100% due to rounding.

Map Projection Distortion

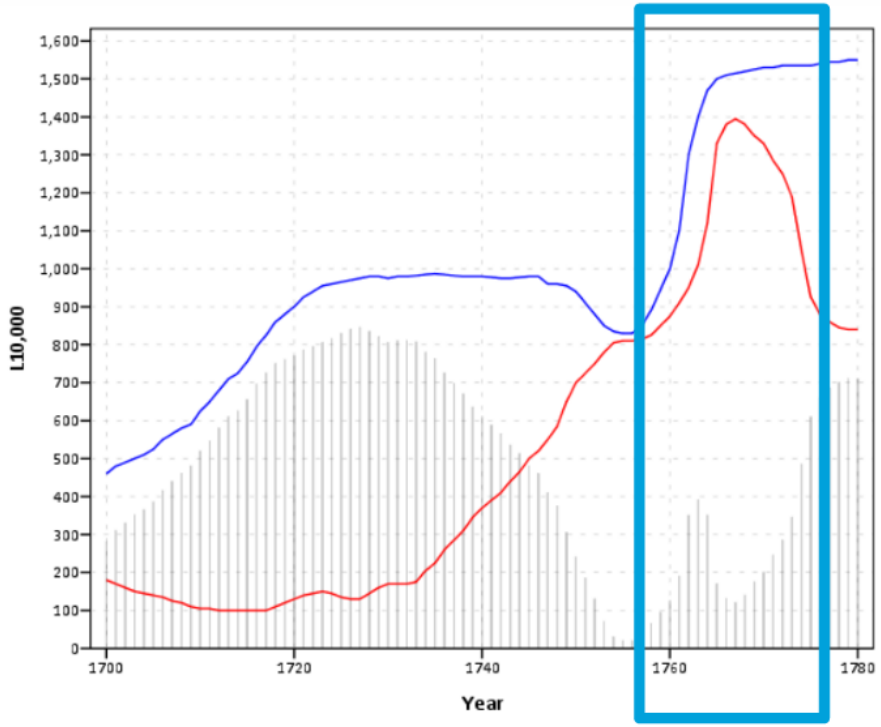


Perception – Visual Illusion

Inappropriate Aspect Ratio



Sine Illusion



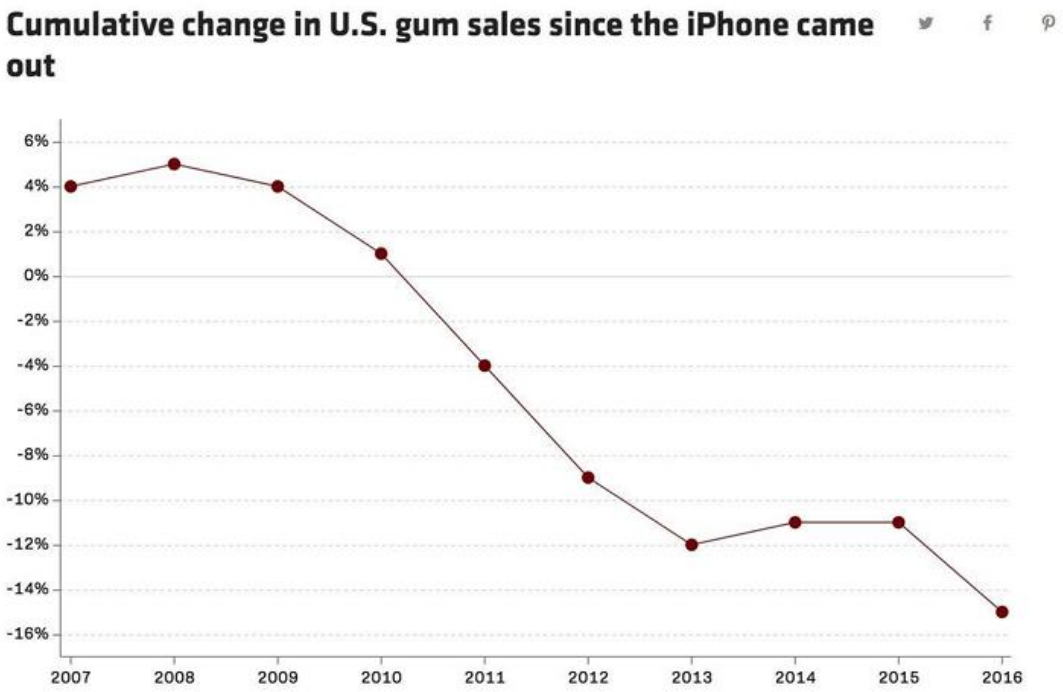
Red line is exports, Blue line is imports, grey bars are Imports minus exports.

The gray bars are correct, but look wrong

Interpretation – False Linkage

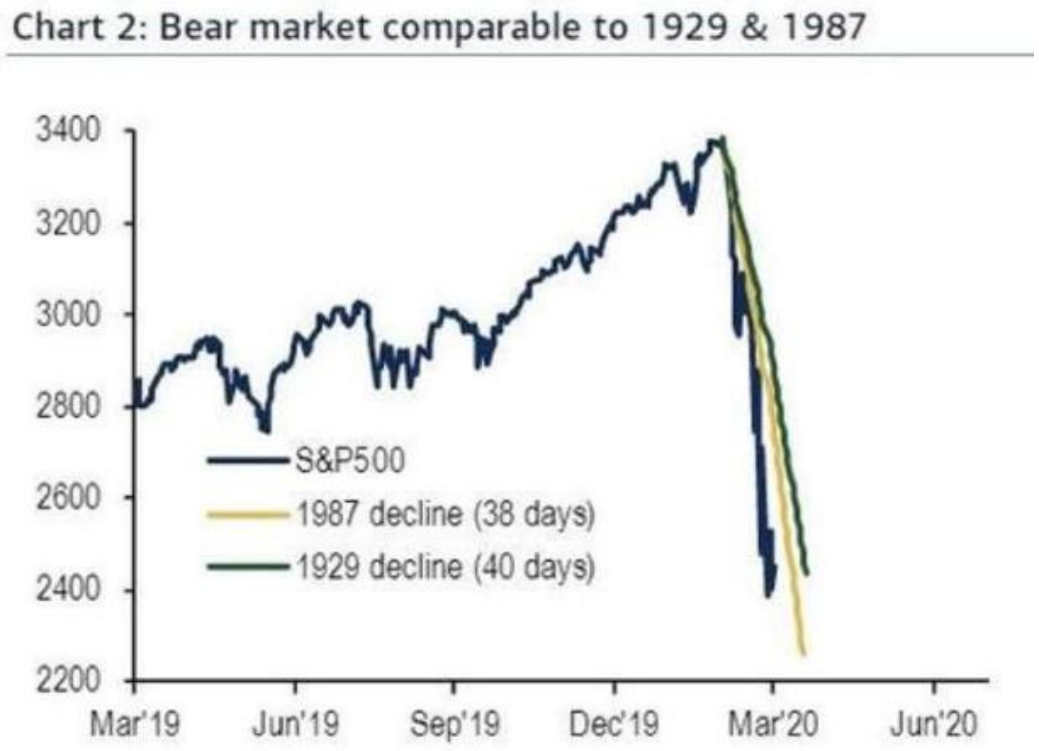
Invalid Comparison

Correlation Not Causation



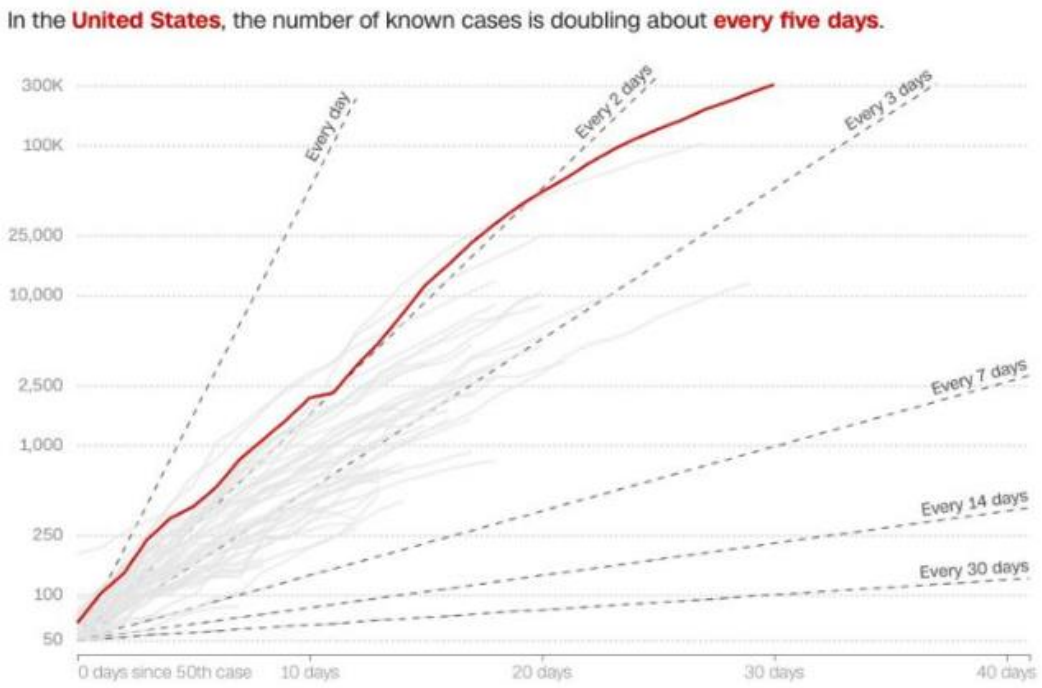
Interpretation – False Linkage

Pattern Seeking

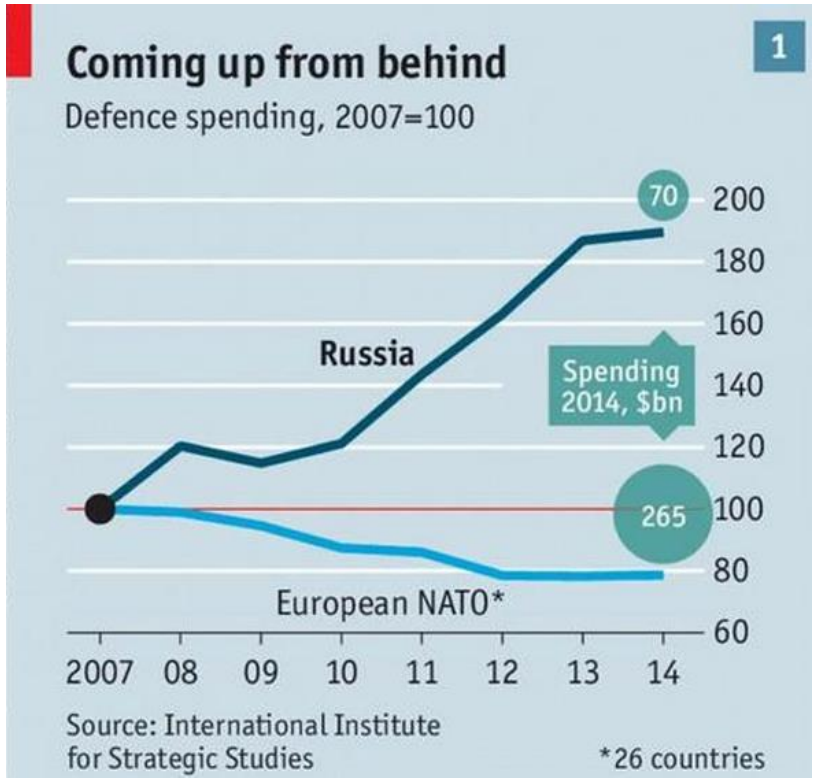


Interpretation – Power of Words

Misleading Claim

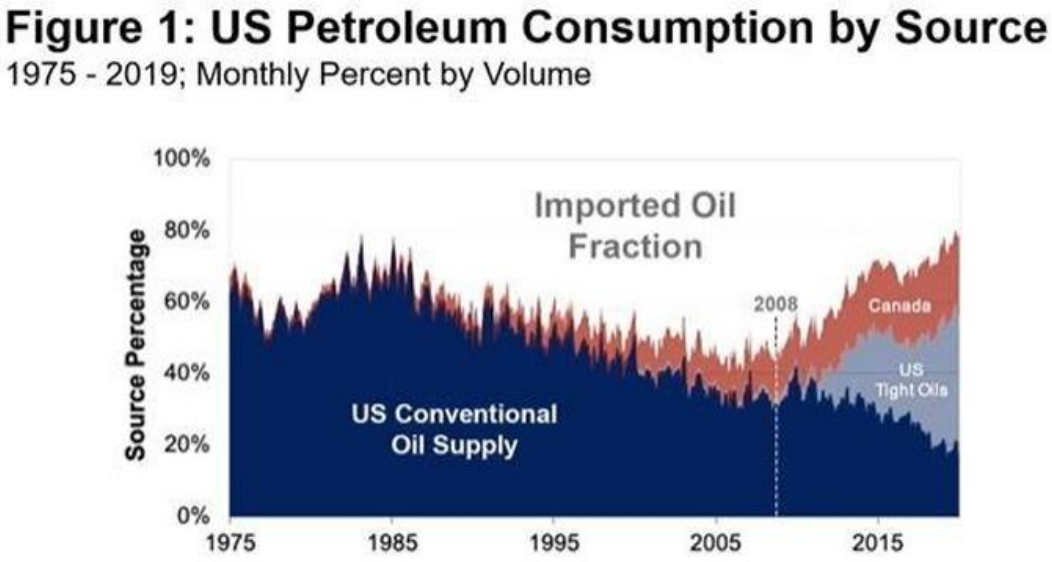


Misleading Annotation

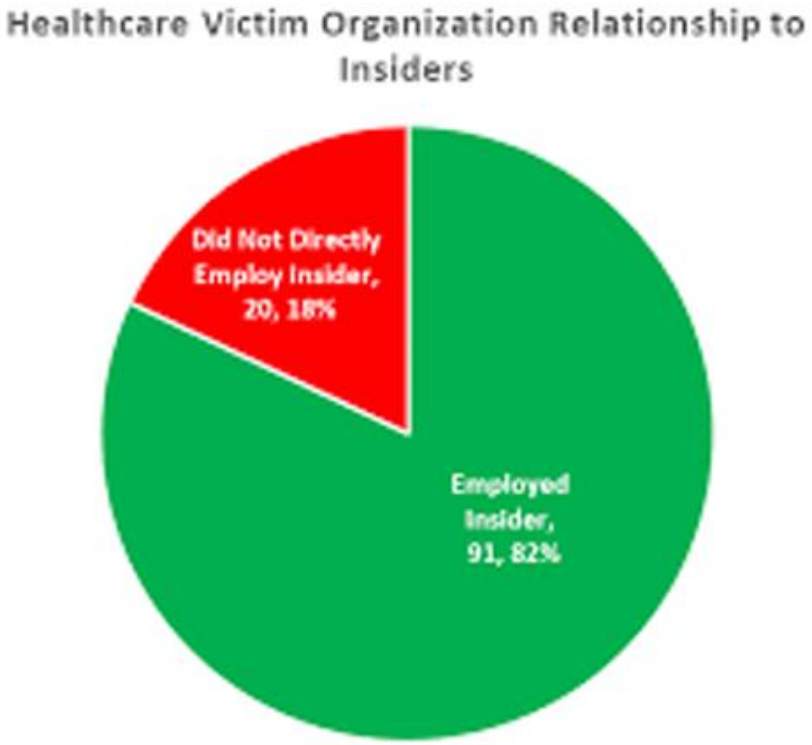


Interpretation – Power of Words

Misleading Title

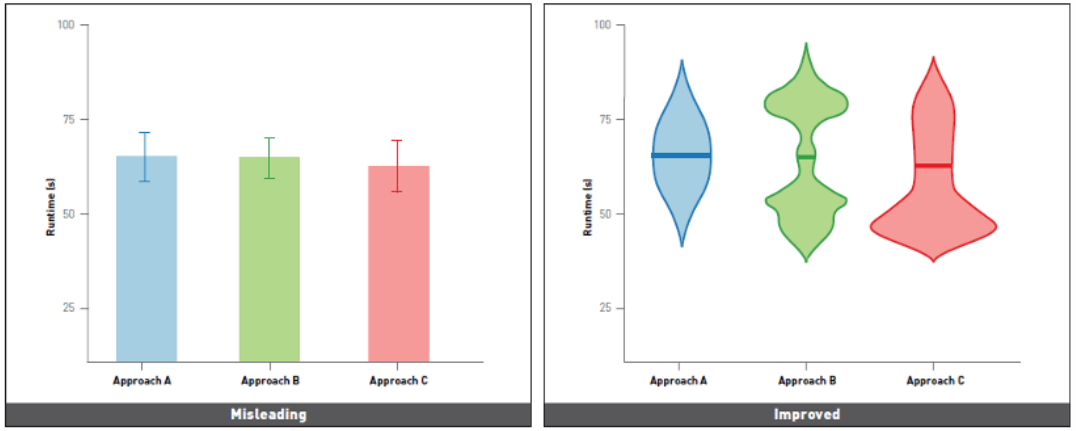


Misleading Value Labels

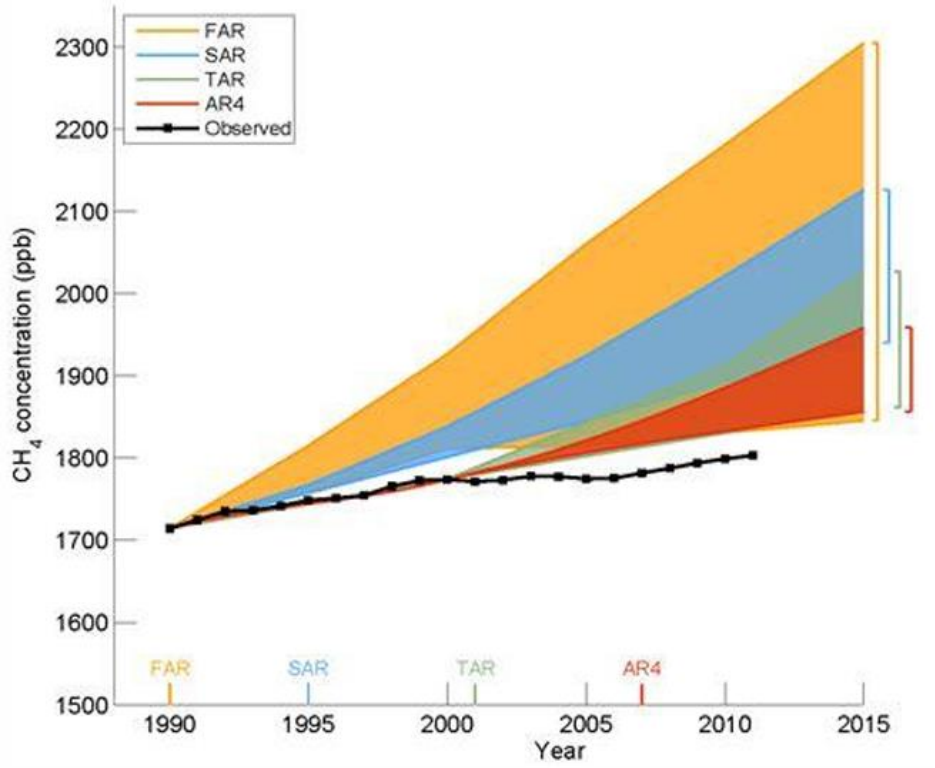


Interpretation – Hidden Information

Hidden Distribution

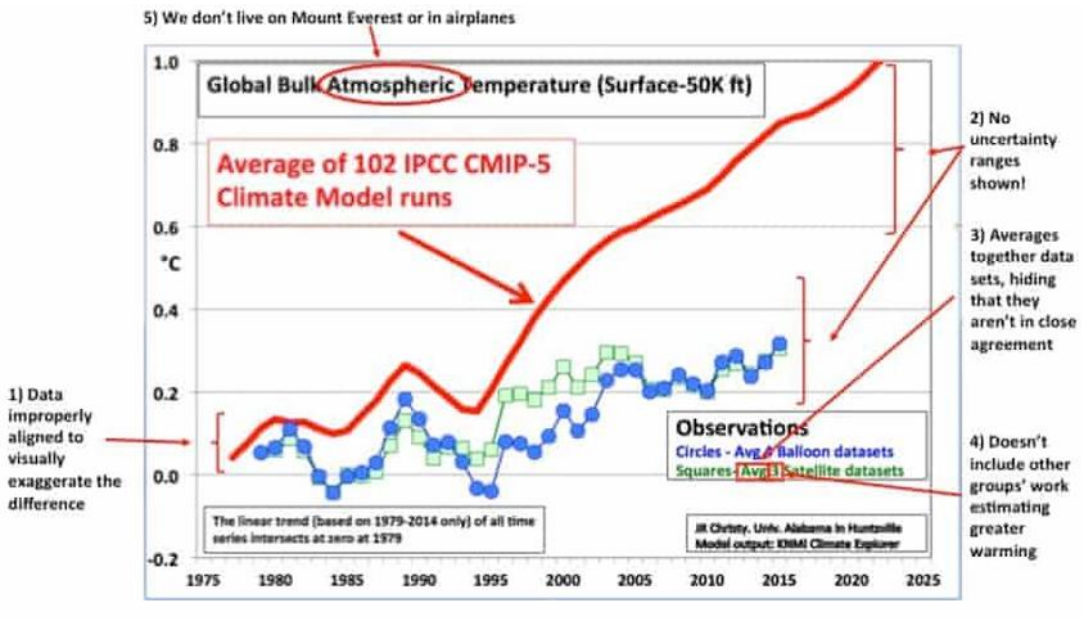
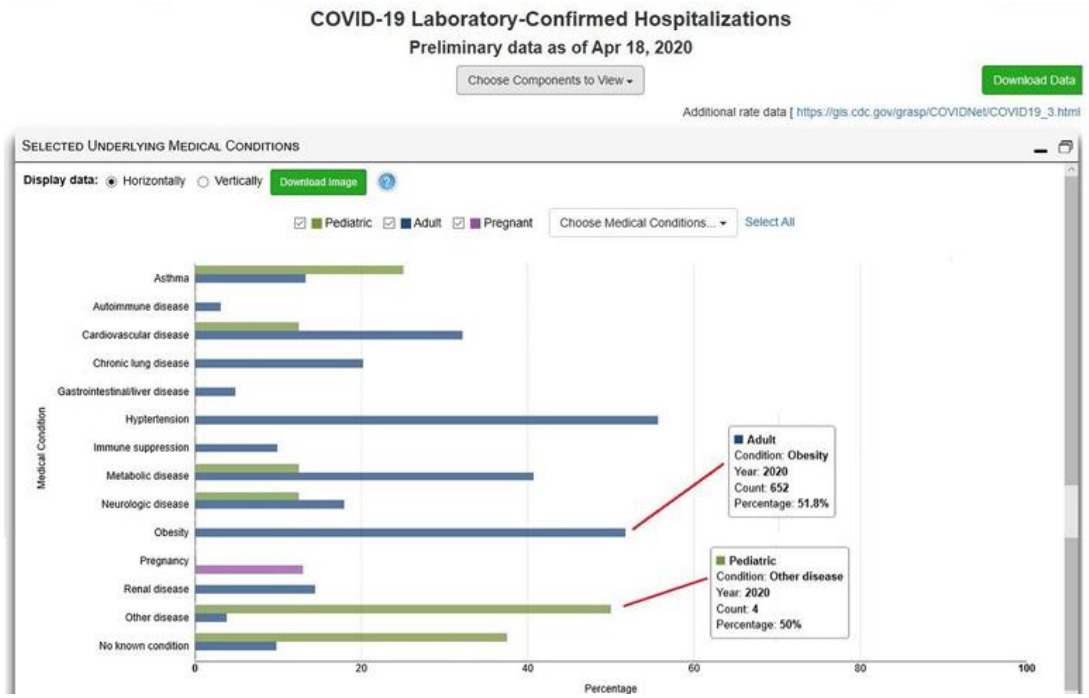


Overplotting



Interpretation – Hidden Information

Hidden Uncertainty



Hidden Population Size